



Light-Sensitive Switch Using an LDR Voltage Divider

Design, Calculation, Proteus Simulation,
Troubleshooting, and Project Questions

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Project Report

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Abstract

This project presents the design, calculation, simulation, troubleshooting, and review of a light-sensitive switch circuit based on a Light Dependent Resistor and a fixed-resistor voltage divider. The circuit uses a 5 V direct current supply, an LDR connected above the output node, and a 10 k Ω resistor connected between the output node and ground. The output voltage is measured at V_{out} and changes according to the light received by the LDR. When light intensity increases, the LDR resistance decreases, the voltage drop across the LDR becomes smaller, and the output voltage rises. The report explains the working principle, derives the voltage-divider equation, calculates expected values for dark, medium-light, and bright-light conditions, verifies the circuit arrangement in Proteus, compares theoretical and simulated results, and identifies common troubleshooting cases. The project demonstrates how a simple sensor circuit can convert an environmental condition into a measurable electrical signal suitable for further control or switching applications.

Keywords: Light Dependent Resistor, voltage divider, light-sensitive switch, Proteus simulation, troubleshooting.

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1

Description

1.1 Introduction

A light-sensitive switch is an electronic circuit designed to respond to changes in light intensity. The central purpose of this project is to demonstrate how environmental light can be converted into an electrical signal by using a Light Dependent Resistor and a voltage divider arrangement. The circuit does not treat light as an abstract condition, but as a physical input that changes the resistance of the sensing component. This resistance change is then transformed into a measurable voltage variation at the output node of the circuit.

The project is based on a simple but important electronic principle. When two resistive elements are connected in series across a direct current supply, the voltage at the junction between them depends on the ratio of their resistances. In this circuit, one of the resistive elements is fixed, while the other changes according to light intensity. As a result, the output voltage becomes dependent on the amount of light falling on the Light Dependent Resistor.

The circuit is designed and tested in Proteus simulation software. The use of simulation allows the circuit behavior to be observed before physical construction. By moving the virtual torch closer to or farther from the Light Dependent Resistor, the output voltage can be measured with a voltmeter and compared with the expected theoretical behavior. This makes the project suitable for understanding sensor based circuits, voltage division, and basic light controlled switching behavior.

1.2 Project Overview

This project presents a light-sensitive circuit using a Light Dependent Resistor and a fixed resistor arranged as a voltage divider. The supply voltage is 5 volts direct current. The Light Dependent Resistor is connected between the positive supply and the output junction. The fixed resistor, with a value of 10 kilohms, is connected between the output junction and ground. A direct current voltmeter is connected between the output junction and ground to measure the output voltage.

The output node of the circuit is identified as (V_{out}). This voltage changes according to the resistance of the Light Dependent Resistor. In darkness, the resistance of the

Light Dependent Resistor is high, causing the output voltage to be low. In bright light, the resistance of the Light Dependent Resistor becomes low, causing the output voltage to rise. Therefore, the circuit converts light intensity into a voltage level.

The circuit represents the sensing stage of a light-sensitive switch. It provides a variable voltage that can later be used to control another device, such as a transistor, comparator, relay, light emitting diode, or microcontroller input. In its present form, the project focuses on the fundamental sensing and voltage conversion process.

1.3 Aim of the Project

The aim of this project is to design, simulate, and analyze a light-sensitive voltage divider circuit in which the output voltage changes according to the intensity of light received by a Light Dependent Resistor.

More specifically, the project aims to show that when the Light Dependent Resistor is connected above the output node and the fixed resistor is connected below the output node, the output voltage increases as light intensity increases. This behavior is not random. It follows directly from the voltage divider equation and from the physical behavior of the Light Dependent Resistor.

The central aim can be stated as follows:

To convert a change in light intensity into a measurable change in output voltage.

This aim is achieved through the relationship between the supply voltage, the Light Dependent Resistor resistance, the fixed resistor value, and the output voltage.

1.4 Objectives of the Project

The objectives of the project are to establish both theoretical understanding and practical simulation verification.

The first objective is to understand the behavior of a Light Dependent Resistor. The Light Dependent Resistor changes its resistance according to the amount of light falling on its surface. Higher light intensity produces lower resistance, while lower light intensity produces higher resistance.

The second objective is to apply the voltage divider principle. The project demonstrates how two resistive elements connected in series can divide the supply voltage and produce an output voltage at their junction.

The third objective is to construct the correct circuit arrangement in Proteus. This includes placing the 5 volt direct current supply, the Light Dependent Resistor, the fixed resistor, the ground terminal, the voltmeter, and the virtual torch in the correct positions.

The fourth objective is to measure the output voltage at the junction between the Light Dependent Resistor and the fixed resistor. The voltmeter must be connected with its positive terminal at (V_{out}) and its negative terminal at ground.

The fifth objective is to observe how the output voltage changes when the virtual torch is moved closer to the Light Dependent Resistor. The expected result is that the voltmeter reading increases when light intensity increases.

The sixth objective is to verify that the simulation result agrees with the mathematical voltage divider model. This objective connects the theoretical calculation with the observed Proteus output.

The seventh objective is to identify common faults that may occur in the circuit, including incorrect wiring, wrong resistor values, missing ground, incorrect voltmeter polarity, and inverted sensor response.

1.5 Background of the Light-Sensitive Switch

A light-sensitive switch is a circuit that changes its electrical state or output condition according to light intensity. Such circuits are commonly used in automatic lighting systems, street lights, alarm systems, brightness detection circuits, and sensor based control systems. The main idea behind these circuits is that the environment can influence an electrical parameter, and that electrical parameter can then be used for measurement or control.

The Light Dependent Resistor is one of the simplest components used for light detection. It behaves as a variable resistor whose resistance depends on illumination. In low light, its resistance is large. In strong light, its resistance becomes small. This property allows it to serve as a basic light sensor.

However, the Light Dependent Resistor alone does not directly provide a useful output voltage. It must be placed in a circuit that converts resistance variation into voltage variation. The voltage divider performs this conversion. By combining the Light Dependent Resistor with a fixed resistor, the circuit creates an output voltage that changes predictably with light.

This project therefore belongs to the foundation of sensor electronics. It shows how a physical condition, namely light intensity, can be represented as an electrical signal. This is an essential concept in electronic measurement, automation, and control.

1.6 Function of the LDR

The Light Dependent Resistor is the sensing component of the circuit. Its main function is to detect the intensity of light and change its resistance accordingly.

When little or no light falls on the Light Dependent Resistor, its resistance is high. This high resistance limits the flow of current through the upper part of the voltage divider. Since the Light Dependent Resistor is connected between the positive supply and the output node, a high resistance causes a large voltage drop across the Light Dependent Resistor, leaving a smaller voltage at the output node.

When strong light falls on the Light Dependent Resistor, its resistance decreases. This lower resistance reduces the voltage drop across the Light Dependent Resistor. As a result, the output node receives a higher voltage.

The behavior can be summarized as follows:

Low light $\Rightarrow R_{LDR}$ increases

$\Rightarrow V_{out}$ decreases

High light $\Rightarrow R_{LDR}$ decreases

$\Rightarrow V_{out}$ increases

The Light Dependent Resistor therefore acts as the environmental input device of the circuit.

1.7 Function of the Voltage Divider

The voltage divider is the mathematical and electrical foundation of the project. It consists of two resistive elements connected in series across a supply voltage. The output voltage is taken from the junction between the two elements.

In this circuit, the upper resistive element is the Light Dependent Resistor, and the lower resistive element is the fixed 10 kilohm resistor. The output voltage is measured between their junction and ground.

The voltage divider equation for this circuit is:

$$V_{out} = V_{CC} \times \frac{R_1}{R_{LDR} + R_1}$$

where (V_{out}) is the output voltage, (V_{CC}) is the supply voltage, (R_{LDR}) is the resistance of the Light Dependent Resistor, and (R_1) is the fixed resistor.

Since the supply voltage is 5 volts and the fixed resistor is 10 kilohms, the equation becomes:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This formula proves the expected behavior of the circuit. If (R_{LDR}) decreases because of stronger light, the denominator becomes smaller. Since the numerator remains constant, the output voltage increases. Therefore, the voltage divider transforms the changing resistance of the Light Dependent Resistor into a changing voltage.

1.8 Main Components Used

The main components used in this project are the 5 volt direct current supply, the Light Dependent Resistor, the fixed resistor, the direct current voltmeter, the ground terminal, and the virtual torch.

The 5 volt direct current supply provides the electrical energy required for the circuit. Its positive terminal is connected to the upper side of the Light Dependent Resistor, and its negative terminal is connected to ground.

The Light Dependent Resistor is the sensor of the circuit. It changes resistance according to the light intensity received from the virtual torch.

The fixed resistor has a value of 10 kilohms. It is connected between the output node and ground. Its function is to complete the voltage divider and provide a stable lower resistance path.

The direct current voltmeter measures the voltage at the output node. Its positive lead is connected to (V_{out}), and its negative lead is connected to ground.

The ground terminal provides the zero volt reference point for the entire circuit. Without a correct ground connection, the voltage measurement would not have a stable reference.

The virtual torch is used in Proteus to simulate changing light intensity. When the torch is moved closer to the Light Dependent Resistor, the simulated light intensity increases, causing the resistance of the Light Dependent Resistor to decrease.

1.9 Working Principle of the Circuit

The working principle of the circuit is based on the interaction between light intensity, Light Dependent Resistor resistance, and voltage divider output.

The circuit is supplied by a 5 volt direct current source. The Light Dependent Resistor is connected from the positive supply to the output junction. The fixed 10 kilohm resistor is connected from the output junction to ground. The output voltage is measured at the junction between these two components.

When the circuit is in low light, the Light Dependent Resistor has a high resistance. Because this resistance is located above the output node, a larger portion of the supply voltage is dropped across the Light Dependent Resistor. The remaining voltage at (V_{out}) is therefore small.

When the circuit is exposed to bright light, the resistance of the Light Dependent Resistor decreases. A smaller portion of the supply voltage is then dropped across the Light Dependent Resistor. Consequently, the voltage at the output node increases.

The equation governing the circuit is:

$$V_{out} = V_{CC} \times \frac{R_1}{R_{LDR} + R_1}$$

For the selected values:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

If the Light Dependent Resistor is assumed to have a resistance of 100 kilohms in darkness, then:

$$V_{out} = 5V \times \frac{10k\Omega}{100k\Omega + 10k\Omega}$$

$$V_{out} \approx 0.45V$$

If the Light Dependent Resistor is assumed to have a resistance of 1 kilohm in bright light, then:

$$V_{out} = 5V \times \frac{10k\Omega}{1k\Omega + 10k\Omega}$$

$$V_{out} \approx 4.55V$$

These values show that the circuit produces a low voltage in darkness and a high voltage in bright light. This is the required behavior for a light-sensitive sensing circuit.

1.10 Expected Output Behavior

The expected output behavior is that the voltage at (V_{out}) increases when light intensity increases and decreases when light intensity decreases.

In darkness, the resistance of the Light Dependent Resistor is high. This causes the output voltage to remain low. Under medium light, the resistance of the Light Dependent Resistor decreases to a moderate value, and the output voltage moves toward the middle of the supply range. Under bright light, the resistance of the Light Dependent Resistor becomes low, and the output voltage rises close to the supply voltage.

The expected relationship can be expressed as:

Dark condition $\Rightarrow R_{LDR}$ high
 $\Rightarrow V_{out}$ low
 Medium light condition $\Rightarrow R_{LDR}$ medium
 $\Rightarrow V_{out}$ medium
 Bright light condition $\Rightarrow R_{LDR}$ low
 $\Rightarrow V_{out}$ high

A typical set of expected values is shown below.

Table 1.1: *Expected output voltage under different light conditions*

Light condition	Approximate Light Dependent Resistor resistance	Expected output voltage
Darkness	100 kilohms	0.45 volts
Medium light	10 kilohms	2.50 volts
Bright light	1 kilohm	4.55 volts

This expected behavior confirms that the circuit is suitable for detecting changes in light intensity.

1.11 Importance of the Project

This project is important because it introduces a fundamental method used in electronic sensing systems. Many practical electronic devices depend on the conversion of physical changes into electrical signals. In this project, light intensity is the physical condition, resistance is the intermediate electrical property, and voltage is the measurable output.

The project also strengthens understanding of voltage division. The voltage divider is one of the most common and essential circuit arrangements in electronics. It is used in sensors, biasing networks, reference voltage circuits, analog measurement systems, and input conditioning circuits.

The project is also important because it connects theory with simulation. The mathematical equation predicts the output voltage, while the Proteus simulation allows the result to be observed visually and numerically. This connection between calculation and simulation improves technical accuracy and supports engineering reasoning.

Finally, the project provides a foundation for more advanced light-controlled systems. The output voltage from this circuit can be connected to a comparator, transistor, relay driver, or microcontroller. Therefore, the circuit is not only a small experiment, but also the first stage of a larger automatic control system.

1.12 Description Image Placement

The description image should be placed after the explanation of the working principle and before the chapter summary. This position is appropriate because the reader first receives the conceptual explanation, then sees the visual representation of the circuit.

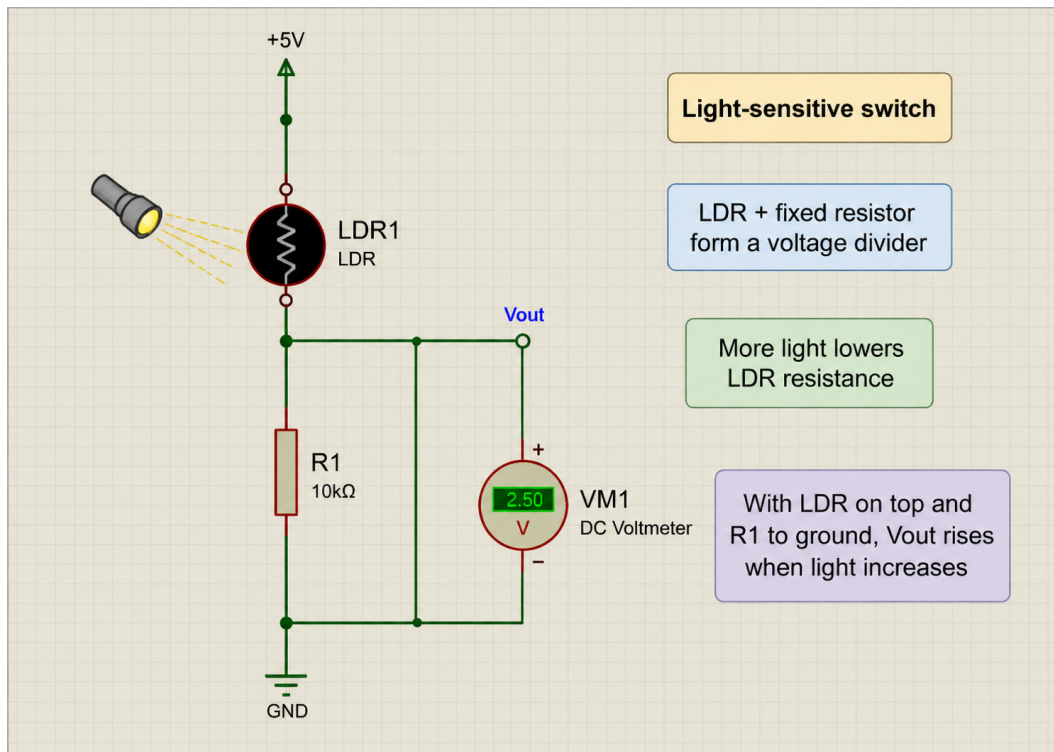


Figure 1.1: *Light-sensitive switch concept diagram*

The figure should show the full circuit arrangement clearly. The Light Dependent Resistor should be connected between the positive 5 volt supply and the output node. The fixed 10 kilohm resistor should be connected between the output node and ground. The voltmeter should be connected between (V_{out}) and ground. The virtual torch should be directed toward the Light Dependent Resistor.

The image should contain the following labels:

First, the positive supply should be labeled as (+5V). Second, the Light Dependent Resistor should be labeled as (LDR1). Third, the fixed resistor should be labeled as ($R1 = 10k\Omega$). Fourth, the output node should be labeled as (V_{out}). Fifth, the ground reference should be labeled as (GND). Sixth, the measuring instrument should be labeled as (VM1).

A suitable figure title is:

Figure 1.1: Light-sensitive voltage divider circuit using a Light Dependent Resistor

The figure should support the text by making the circuit structure immediately understandable. It must not replace the explanation. Instead, it should confirm visually what the written description has already established.

1.13 Chapter Summary

This chapter introduced the light-sensitive switch project based on a Light Dependent Resistor and a voltage divider circuit. The project uses a 5 volt direct current supply, a Light Dependent Resistor, a 10 kilohm fixed resistor, a voltmeter, a ground terminal, and a virtual torch in Proteus.

The chapter explained that the Light Dependent Resistor changes resistance according to light intensity. In darkness, its resistance is high. In bright light, its resistance is low. The voltage divider converts this resistance change into a voltage change at the output node.

The chapter also established the main operating rule of the circuit. Since the Light Dependent Resistor is connected between the positive supply and the output node, and the fixed resistor is connected between the output node and ground, the output voltage increases when light intensity increases. This behavior is supported by the voltage divider equation:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

The chapter therefore provides the conceptual, functional, and mathematical foundation for the rest of the report. Chapter 2 will develop the calculation in greater detail and prove the expected output voltage values under different light conditions.

2

Calculation

2.1 Introduction to Circuit Calculation

This chapter presents the mathematical analysis of the light sensitive switch circuit. The purpose of the calculation is to explain how the Light Dependent Resistor and the fixed resistor produce a measurable output voltage. The circuit is based on a voltage divider arrangement, where the supply voltage is divided between two resistive elements connected in series.

In this project, the Light Dependent Resistor acts as the variable resistance, while the fixed resistor provides a stable reference resistance. The output voltage is taken from the junction between these two components. Since the resistance of the Light Dependent Resistor changes according to the amount of light falling on it, the output voltage also changes. Therefore, the calculation is necessary to show the exact relationship between light intensity, resistance variation, and output voltage.

The mathematical analysis also helps to predict the expected voltmeter reading before running the Proteus simulation. This is important because a simulation result is only meaningful when it can be compared with a theoretical expectation. If the simulated voltage follows the calculated trend, then the circuit behavior is considered correct. If the simulated voltage does not follow the calculated trend, then the circuit must be checked for wiring errors, component value mistakes, or incorrect simulation settings.

The calculation in this chapter is based on a 5 volt direct current supply, a 10 kilo ohm fixed resistor, and an Light Dependent Resistor whose resistance changes between high and low values depending on the light condition.

2.2 Voltage Divider Principle

A voltage divider is a basic electrical circuit that produces an output voltage smaller than or equal to the input voltage. It is formed by connecting two resistors in series across a voltage supply. The output voltage is measured from the junction between the two resistors to the ground reference.

The voltage divider principle is based on the fact that the same current flows through both resistors in a series circuit. However, the supply voltage is shared between the two resistors according to their resistance values. A resistor with a larger resistance receives

a larger share of the voltage, while a resistor with a smaller resistance receives a smaller share of the voltage.

In the light sensitive switch circuit, the two resistive elements are the Light Dependent Resistor and the fixed resistor. The Light Dependent Resistor is connected between the positive supply and the output junction. The fixed resistor is connected between the output junction and ground. The output voltage is measured at the junction between them.

When the Light Dependent Resistor is in darkness, its resistance is high. This causes most of the supply voltage to drop across the Light Dependent Resistor, so the voltage at the output junction becomes low. When the Light Dependent Resistor is exposed to bright light, its resistance becomes low. This reduces the voltage drop across the Light Dependent Resistor, so the output junction voltage becomes high.

Therefore, this circuit converts a change in light intensity into a change in output voltage.

2.3 General Voltage Divider Formula

The general voltage divider formula is:

$$V_{out} = V_{CC} \times \frac{R_{bottom}}{R_{top} + R_{bottom}}$$

In this formula, (V_{out}) is the output voltage measured at the junction between the two resistors. (V_{CC}) is the supply voltage applied across the full voltage divider. (R_{top}) is the resistance connected between the positive supply and the output node. (R_{bottom}) is the resistance connected between the output node and ground.

This formula shows that the output voltage depends on the ratio between the bottom resistance and the total resistance of the divider. The output voltage increases when the bottom resistance becomes large compared with the total resistance. The output voltage decreases when the top resistance becomes large compared with the bottom resistance.

The formula is suitable for this project because the Light Dependent Resistor and the fixed resistor are connected in series, and the output voltage is taken from their junction. As long as the voltmeter has very high internal resistance, it does not significantly disturb the circuit. Therefore, the voltage divider equation can be used to calculate the expected output voltage accurately.

2.4 Identification of Top and Bottom Resistors

The correct identification of the top and bottom resistors is essential because the output direction depends on their positions. In this circuit, the Light Dependent Resistor is the top resistor because it is connected between the positive 5 volt supply and the output junction. The fixed resistor is the bottom resistor because it is connected between the output junction and ground.

Therefore:

$$\begin{aligned} R_{top} &= R_{LDR} \\ R_{bottom} &= R_1 \end{aligned}$$

The fixed resistor has the value:

$$R_1 = 10k\Omega$$

The supply voltage is:

$$V_{CC} = 5V$$

This arrangement is intentionally selected so that the output voltage rises when the light intensity increases. When the Light Dependent Resistor receives more light, its resistance decreases. Since this decreasing resistance is located above the output node, the voltage drop above the output node becomes smaller. As a result, the voltage at the output node becomes higher.

If the positions of the Light Dependent Resistor and the fixed resistor were exchanged, the output behavior would be reversed. In that alternative arrangement, the output voltage would decrease when light intensity increases. Since this project requires the voltage to rise as the torch moves closer to the Light Dependent Resistor, the selected arrangement is correct.

2.5 Circuit Specific Formula

After identifying the top and bottom resistors, the general voltage divider formula can be rewritten for this specific circuit.

The general formula is:

$$V_{out} = V_{CC} \times \frac{R_{bottom}}{R_{top} + R_{bottom}}$$

For this circuit:

$$V_{CC} = 5V$$

$$R_{top} = R_{LDR}$$

$$R_{bottom} = 10k\Omega$$

Substituting these values into the general formula gives:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This is the main calculation formula for the project. It shows that the output voltage is controlled by the resistance of the Light Dependent Resistor. The fixed resistor remains constant, while the Light Dependent Resistor changes according to the light condition.

The formula also proves the expected behavior of the circuit. If (R_{LDR}) becomes very large, the denominator becomes large, and (V_{out}) becomes small. If (R_{LDR}) becomes small, the denominator becomes smaller, and (V_{out}) becomes larger. Therefore, the output voltage increases when light intensity increases.

2.6 Assumed LDR Resistance Values

The resistance of a Light Dependent Resistor is not fixed. It depends on the amount of light that falls on the sensor surface. For calculation purposes, representative resistance values are used to model three operating conditions: dark condition, medium light condition, and bright light condition.

In this report, the following assumed values are used:

Dark condition:

$$R_{LDR} = 100k\Omega$$

Medium light condition:

$$R_{LDR} = 10k\Omega$$

Bright light condition:

$$R_{LDR} = 1k\Omega$$

These values are suitable for explaining the general behavior of the circuit. They show the transition from a high resistance state in darkness to a low resistance state in bright light. The exact values in a real component may vary depending on the specific Light Dependent Resistor model, the light source intensity, and the physical distance between the light source and the sensor. However, the electrical principle remains the same.

The purpose of these assumed values is not to claim that every Light Dependent Resistor will always have exactly these resistances. Their purpose is to demonstrate the voltage divider behavior clearly and to provide a theoretical basis for comparison with simulation results.

2.7 Dark Condition Calculation

In the dark condition, the Light Dependent Resistor has a high resistance. For this calculation, the assumed resistance is:

$$R_{LDR} = 100k\Omega$$

The fixed resistor is:

$$R_1 = 10k\Omega$$

The supply voltage is:

$$V_{CC} = 5V$$

Using the circuit specific formula:

$$V_{out} = 5V \times \frac{10k\Omega}{100k\Omega + 10k\Omega}$$

First, the total resistance is calculated:

$$100k\Omega + 10k\Omega = 110k\Omega$$

Then the resistance ratio is:

$$\frac{10k\Omega}{110k\Omega} = \frac{10}{110}$$

$$\frac{10}{110} = 0.0909$$

Therefore:

$$V_{out} = 5V \times 0.0909$$

$$V_{out} = 0.4545V$$

Rounded to two decimal places:

$$V_{out} \approx 0.45V$$

This result shows that the output voltage is low in darkness. The reason is that the high resistance of the Light Dependent Resistor takes most of the supply voltage, leaving only a small voltage at the output node.

2.8 Medium Light Condition Calculation

In the medium light condition, the Light Dependent Resistor is assumed to have a resistance equal to the fixed resistor. Therefore:

$$R_{LDR} = 10k\Omega$$

The fixed resistor is:

$$R_1 = 10k\Omega$$

The supply voltage is:

$$V_{CC} = 5V$$

Using the circuit specific formula:

$$V_{out} = 5V \times \frac{10k\Omega}{10k\Omega + 10k\Omega}$$

First, the total resistance is calculated:

$$10k\Omega + 10k\Omega = 20k\Omega$$

Then the resistance ratio is:

$$\frac{10k\Omega}{20k\Omega} = \frac{1}{2}$$

$$\frac{1}{2} = 0.5$$

Therefore:

$$V_{out} = 5V \times 0.5$$

$$V_{out} = 2.5V$$

This result shows that when the Light Dependent Resistor and the fixed resistor have equal resistance values, the output voltage is exactly half of the supply voltage. Since the supply voltage is 5 volts, the output voltage becomes 2.5 volts.

This condition is important because it represents the balanced point of the voltage divider. At this point, the voltage is divided equally between the Light Dependent Resistor and the fixed resistor.

2.9 Bright Light Condition Calculation

In the bright light condition, the Light Dependent Resistor has a low resistance. For this calculation, the assumed resistance is:

$$R_{LDR} = 1k\Omega$$

The fixed resistor is:

$$R_1 = 10k\Omega$$

The supply voltage is:

$$V_{CC} = 5V$$

Using the circuit specific formula:

$$V_{out} = 5V \times \frac{10k\Omega}{1k\Omega + 10k\Omega}$$

First, the total resistance is calculated:

$$1k\Omega + 10k\Omega = 11k\Omega$$

Then the resistance ratio is:

$$\frac{10k\Omega}{11k\Omega} = \frac{10}{11}$$

$$\frac{10}{11} = 0.9091$$

Therefore:

$$V_{out} = 5V \times 0.9091$$

$$V_{out} = 4.5455V$$

Rounded to two decimal places:

$$V_{out} \approx 4.55V$$

This result shows that the output voltage is high under bright light. The reason is that the Light Dependent Resistor has low resistance, so only a small part of the supply voltage is dropped across it. Most of the voltage appears across the fixed resistor, and because the output is measured across the fixed resistor relative to ground, the measured output voltage becomes high.

2.10 Output Voltage Comparison Table

The calculated values can be compared in a single table to show the relationship between light condition, Light Dependent Resistor resistance, and output voltage.

Table 2.1: *Calculated output voltage comparison*

Light condition	Assumed Light Dependent Resistor resistance	Fixed resistor value	Calculated output voltage	Output state
Dark	100 kilo ohm	10 kilo ohm	0.45 volt	Low
Medium light	10 kilo ohm	10 kilo ohm	2.50 volt	Medium
Bright light	1 kilo ohm	10 kilo ohm	4.55 volt	High

The table confirms that the output voltage increases as the resistance of the Light Dependent Resistor decreases. This is the expected result for the selected circuit arrangement.

The comparison also shows that the circuit has a clear voltage range. In darkness, the voltage is close to 0 volts. Under medium light, the voltage is around half of the supply voltage. Under bright light, the voltage approaches the supply voltage. This makes the circuit suitable as the sensing stage of a light sensitive switching system.

2.11 Interpretation of Calculation Results

The calculation results prove that the output voltage is directly controlled by the resistance of the Light Dependent Resistor. Since the resistance of the Light Dependent Resistor decreases when light increases, the output voltage increases when light increases.

The dark condition calculation gives approximately 0.45 volts. This means the output is low when the Light Dependent Resistor is not strongly illuminated. The medium condition calculation gives 2.50 volts. This means the output reaches half of the supply voltage when the Light Dependent Resistor and the fixed resistor have equal values. The bright condition calculation gives approximately 4.55 volts. This means the output becomes high when the Light Dependent Resistor receives strong light.

These results also confirm the correctness of placing the Light Dependent Resistor above the output node and the fixed resistor below the output node. In this arrangement, the output voltage rises with increasing light intensity. This matches the intended

behavior of the project, where moving the torch closer to the Light Dependent Resistor should increase the voltmeter reading.

The calculation also provides a standard for judging the simulation. If the Proteus simulation shows a low output voltage in darkness and a high output voltage in bright light, then the simulation agrees with the theoretical model. If the simulation does not show this trend, then the problem is likely caused by wiring, component values, measurement polarity, grounding, or simulation configuration.

2.12 Calculation Image Placement

The calculation image should be placed after the mathematical explanation and before the chapter summary. Its purpose is to visually connect the formula with the circuit structure. The image should show the Light Dependent Resistor connected between the positive 5 volt supply and the output node, the 10 kilo ohm fixed resistor connected between the output node and ground, and the voltmeter connected between the output node and ground.

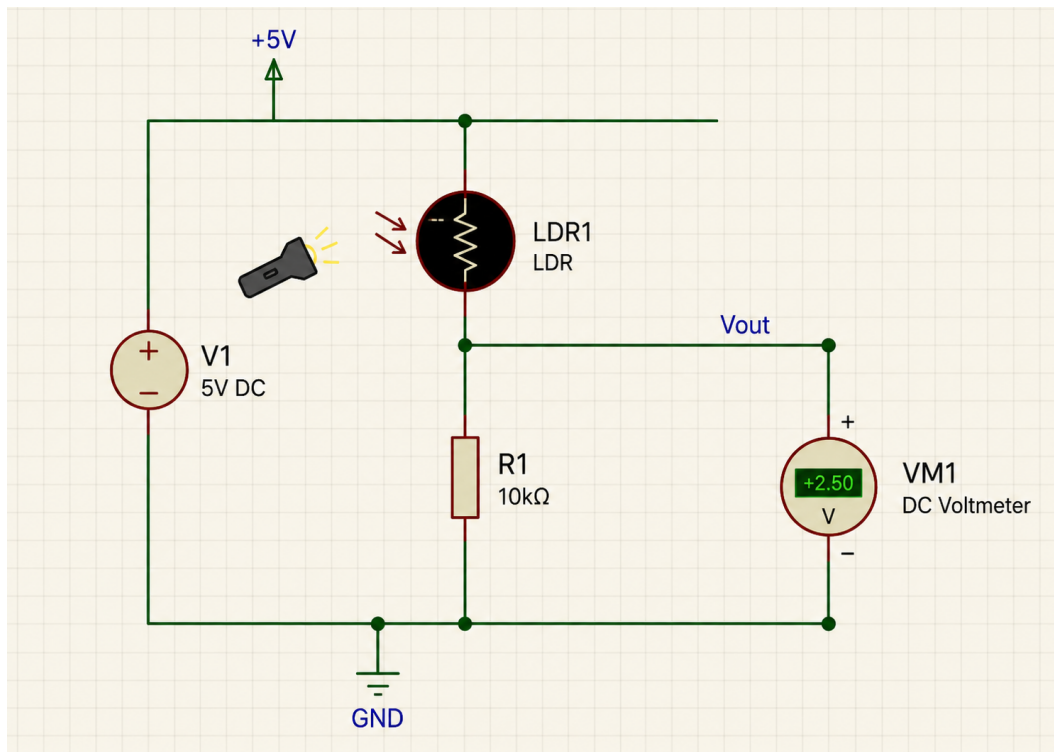


Figure 2.1: Circuit reference used for voltage-divider calculation

The image should also include the circuit specific voltage divider formula:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

The image should include the three calculated operating conditions. In darkness, (R_{LDR}) is 100 kilo ohm and (V_{out}) is approximately 0.45 volt. In medium light, (R_{LDR}) is 10 kilo ohm and (V_{out}) is 2.50 volt. In bright light, (R_{LDR}) is 1 kilo ohm and (V_{out}) is approximately 4.55 volt.

The calculation image should not replace the written explanation. It should support the written explanation by making the mathematical relationship easier to understand. The report should first explain the formula in words, then show the substitutions, then present the calculated values, and finally use the image as visual confirmation.

2.13 Chapter Summary

This chapter calculated the expected output voltage of the light sensitive switch circuit using the voltage divider principle. The Light Dependent Resistor was identified as the top resistor, and the 10 kilo ohm fixed resistor was identified as the bottom resistor. Based on this arrangement, the output voltage was calculated using the formula:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

The dark condition calculation showed that when the Light Dependent Resistor is 100 kilo ohm, the output voltage is approximately 0.45 volt. The medium light calculation showed that when the Light Dependent Resistor is 10 kilo ohm, the output voltage is 2.50 volts. The bright light calculation showed that when the Light Dependent Resistor is 1 kilo ohm, the output voltage is approximately 4.55 volts.

The calculations confirm that the output voltage increases when the light intensity increases. This happens because stronger light reduces the resistance of the Light Dependent Resistor. Since the Light Dependent Resistor is connected above the output node, its resistance decrease causes the output node voltage to rise. Therefore, the mathematical analysis supports the intended behavior of the circuit and prepares the foundation for the Proteus scheme and simulation chapters.

3

Scheme

3.1 Introduction to the Circuit Scheme

This chapter presents the complete circuit scheme of the light sensitive switch based on an LDR and a voltage divider. The purpose of the scheme is to translate the theoretical voltage divider concept into an exact Proteus circuit arrangement. In this chapter, every component, terminal, wire, value, label, and measurement point is defined clearly so that the circuit can be reconstructed without ambiguity.

The circuit is based on a simple but precise arrangement. A light dependent resistor is connected between the positive supply and the output node, while a fixed resistor is connected between the output node and ground. The output voltage is measured at the junction between these two resistive elements. This junction voltage changes according to the resistance of the LDR, which changes according to the amount of light falling on it.

The scheme is therefore not only a visual drawing. It is the formal electrical structure of the project. It identifies how the circuit receives power, how the LDR senses light, how the voltage divider produces an output, how the voltmeter measures the output, and how the ground reference completes the circuit.

3.2 Proteus Components Required

The Proteus scheme requires six main elements. These elements are the DC voltage source, the light dependent resistor, the fixed resistor, the DC voltmeter, the ground terminal, and the virtual torch. Each component has a specific role in the circuit and must be selected correctly.

The DC voltage source provides the electrical energy for the circuit. In this project, the supply voltage is set to 5 V. This value is suitable because it is a common low voltage level used in basic electronics and sensor circuits.

The LDR is the sensing component of the scheme. Its resistance is high when the light level is low, and its resistance decreases when the light level increases. This resistance variation is the main input condition of the circuit.

The fixed resistor is connected with the LDR to form the voltage divider. In this scheme, the fixed resistor has a value of 10 k Ω . This value provides a practical reference

resistance for observing a visible output voltage change across different light conditions.

The DC voltmeter is used to measure the output voltage of the divider. It is connected between the output node and ground. The virtual torch is used as the light source in the simulation environment. It changes the light intensity applied to the LDR and allows the output voltage response to be observed.

3.3 Component Labels and Values

Each component in the circuit must have a clear label and a correct value. The voltage source is labelled V1 and is set to 5 V DC. The light dependent resistor is labelled LDR1. The fixed resistor is labelled R1 and is assigned a value of 10 k Ω . The voltmeter is labelled VM1 and is used as a DC voltage measuring instrument. The ground terminal is labelled GND and represents the zero volt reference of the circuit.

The output node is labelled Vout. This label is placed at the junction between LDR1 and R1. The label is essential because it identifies the exact point where the output voltage is produced and measured. Without this label, the circuit may still work electrically, but the report would not be sufficiently clear for academic explanation.

The component values used in this scheme are as follows. V1 is 5 V DC. R1 is 10 k Ω . LDR1 is treated as a variable resistance controlled by light intensity. VM1 measures the voltage between Vout and GND. The torch has no electrical connection to the circuit because it acts as a simulation light source rather than as an electrical component.

3.4 Circuit Layout Design

The circuit layout should be arranged in a clean vertical voltage divider structure. The positive supply should be placed at the top of the scheme. The LDR should be placed below the positive supply. The output node should be placed directly below the LDR. The fixed resistor should be placed below the output node. The ground terminal should be placed at the bottom of the circuit.

This top to bottom arrangement is important because it visually represents the electrical order of the voltage divider. The upper element is the LDR, the middle point is Vout, and the lower element is the fixed resistor. The output voltage is therefore taken from the exact midpoint between the two resistive elements.

The voltmeter should be placed to the right side of the divider. Its positive terminal should be connected to Vout, and its negative terminal should be connected to GND. This placement allows the measurement path to be visually separate from the main divider path while still being electrically connected to the correct nodes.

The torch should be placed near the LDR and directed toward it. Its position must make clear that light is applied to the LDR, not to the resistor, the voltmeter, or the voltage source.

3.5 Exact Wiring Connections

The first connection is from the positive terminal of V1 to the top terminal of LDR1. This wire establishes the upper supply path of the voltage divider.

The second connection is from the bottom terminal of LDR1 to the Vout node. This point is the central junction of the voltage divider and is the most important node in the circuit.

The third connection is from the Vout node to the top terminal of R1. This connection places the fixed resistor directly below the output node.

The fourth connection is from the bottom terminal of R1 to GND. This wire completes the lower side of the voltage divider and connects the fixed resistor to the reference point of the circuit.

The fifth connection is from the negative terminal of V1 to GND. This connection ensures that the voltage source and the rest of the circuit share the same zero volt reference.

The sixth connection is from the positive terminal of VM1 to Vout. This connection allows the voltmeter to measure the output voltage produced at the junction between LDR1 and R1.

The seventh connection is from the negative terminal of VM1 to GND. This connection allows the voltmeter to measure Vout relative to the reference node.

These connections define the complete electrical structure of the circuit. If any one of these connections is missing, placed incorrectly, or connected to the wrong terminal, the circuit behavior may become invalid.

3.6 Power Supply Connection

The power supply used in the scheme is a 5 V DC source. The positive terminal of the source is connected to the top side of the LDR. This connection provides the input voltage for the voltage divider.

The negative terminal of the source is connected to ground. This is necessary because all voltage measurements in the circuit must be made relative to a common reference point. Without a ground connection, the circuit may have undefined node voltages in the simulation.

The supply voltage is represented as (V_{CC}). In this project, the value of (V_{CC}) is 5 V. The maximum possible output voltage cannot exceed the supply voltage. Therefore, the output voltage must remain between 0 V and 5 V under normal operation.

The supply connection is correct only when the positive side of V1 reaches the upper terminal of LDR1 and the negative side of V1 reaches GND. This creates a complete path from the supply, through the voltage divider, and back to the source reference.

3.7 LDR Connection

The LDR is connected as the upper resistor of the voltage divider. Its top terminal is connected to the positive 5 V supply, and its bottom terminal is connected to the Vout node. This position is deliberate because it makes the output voltage increase when the LDR receives more light.

The LDR resistance is represented as (R_{LDR}). When the light intensity is low, (R_{LDR}) is high. When the light intensity is high, (R_{LDR}) becomes low. Since the LDR is placed above the output node, a lower LDR resistance causes a smaller voltage drop across the LDR and a larger voltage at Vout.

The relationship can be expressed by the circuit formula:

$$V_{out} = V_{CC} \times \frac{R_1}{R_{LDR} + R_1}$$

Since ($V_{CC} = 5V$) and ($R_1 = 10k\Omega$), the circuit specific equation is:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This equation shows why the position of the LDR is important. If the LDR were placed below the output node instead, the output behavior would be reversed.

3.8 Fixed Resistor Connection

The fixed resistor R_1 is connected as the lower resistor of the voltage divider. Its top terminal is connected to V_{out} , and its bottom terminal is connected to GND. The selected value of R_1 is $10k\Omega$.

The role of R_1 is to provide a stable lower resistance path from the output node to ground. Together with the LDR, it creates a voltage division ratio. Without R_1 , the output node would not form a proper divider, and the voltage would not represent a controlled relationship between light intensity and resistance.

The value of $10k\Omega$ is suitable because it is close to the common mid range resistance of many LDR conditions. When (R_{LDR}) becomes equal to $10k\Omega$, the output voltage becomes approximately half of the supply voltage:

$$\begin{aligned} V_{out} &= 5V \times \frac{10k\Omega}{10k\Omega + 10k\Omega} \\ V_{out} &= 2.5V \end{aligned}$$

This midpoint condition is useful because it makes the circuit easy to observe in simulation. A lower LDR resistance produces a higher V_{out} , while a higher LDR resistance produces a lower V_{out} .

3.9 Output Node Connection

The output node is the junction between LDR1 and R_1 . It is labelled V_{out} . This node is the central measurement point of the entire circuit.

Electrically, V_{out} is connected to three points. It is connected to the bottom terminal of LDR1, the top terminal of R_1 , and the positive terminal of VM1. These three connections must meet at the same node. They must not be separated by gaps or placed on different wires.

The output voltage at this node is determined by the ratio between (R_{LDR}) and (R_1). If (R_{LDR}) is large, most of the supply voltage is dropped across the LDR, and V_{out} becomes low. If (R_{LDR}) is small, less voltage is dropped across the LDR, and V_{out} becomes high.

The V_{out} label must be placed directly at the junction. This makes the scheme clear and prevents confusion between the output node, the supply node, and the ground node.

3.10 Voltmeter Connection

The voltmeter VM1 is connected across Vout and GND. The positive terminal of VM1 is connected to Vout. The negative terminal of VM1 is connected to ground. This connection allows the voltmeter to display the output voltage of the divider relative to the circuit reference.

The voltmeter must not be inserted in series with the LDR or the resistor. It must be connected in parallel between the output node and ground. This is because a voltmeter measures potential difference between two points. It does not need to interrupt the main current path.

If the voltmeter leads are reversed, the reading may appear negative. If the positive lead is not connected to Vout, the reading may become zero or incorrect. If the negative lead is not connected to GND, the measurement reference becomes wrong.

The correct voltmeter connection is therefore essential for valid simulation results. The circuit may be physically wired correctly, but the result may still appear wrong if the voltmeter is placed incorrectly.

3.11 Ground Connection

The ground terminal is the zero volt reference of the circuit. It is connected to the bottom terminal of R1, the negative terminal of V1, and the negative terminal of VM1. These points must all share the same ground node.

Ground is necessary because voltage is always measured between two points. In this scheme, Vout is measured relative to GND. Therefore, the ground node defines the reference from which the output voltage is calculated and displayed.

If the ground terminal is missing, the circuit may not simulate correctly. Proteus requires a reference node for circuit analysis. Without ground, the software may treat the circuit as floating, which can cause invalid or unstable readings.

The ground connection also completes the electrical path. Current can flow from the positive supply, through the LDR, through R1, and back to the negative side of the source through ground.

3.12 Torch Placement in Proteus

The torch is placed near the LDR and directed toward the sensing surface of the LDR. The torch is not part of the electrical wiring. Its role is to control the simulated light level applied to the LDR.

When the torch is moved closer to the LDR, the light intensity increases. This causes the LDR resistance to decrease. As a result, the output voltage at Vout increases. When the torch is moved farther away, the light intensity decreases. This causes the LDR resistance to increase, and the output voltage decreases.

The torch should be placed so that its direction is visually clear. It must appear to illuminate the LDR, not another component. This is important for report clarity because the reader must understand that the changing voltage is caused by changing light intensity.

The simulation should use the torch as the controlled environmental input. The voltmeter reading then becomes the observed electrical output.

3.13 Net Labels Used in the Scheme

The scheme uses three essential net labels. These are +5V, Vout, and GND.

The +5V label identifies the positive supply node. It is connected to the positive side of V1 and to the upper terminal of LDR1.

The Vout label identifies the output node of the voltage divider. It is connected to the lower terminal of LDR1, the upper terminal of R1, and the positive terminal of VM1.

The GND label identifies the reference node. It is connected to the bottom terminal of R1, the negative side of V1, and the negative terminal of VM1.

These labels are not decorative. They define the electrical meaning of the circuit. A correctly labelled scheme allows the reader to understand the circuit without tracing every wire manually. It also prevents confusion during simulation, measurement, and troubleshooting.

3.14 Final Circuit Scheme Verification

The final scheme is considered correct only when every component is connected according to its required function. The supply must be 5 V DC. The LDR must be connected between +5V and Vout. The resistor must be connected between Vout and GND. The voltmeter must measure between Vout and GND. The torch must be aimed at the LDR.

The circuit must also satisfy the expected mathematical relationship:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This equation must agree with the wiring. Since R1 is the lower resistor, it appears in the numerator. Since the LDR and R1 form the complete divider resistance, both appear in the denominator.

The final verification must confirm that no terminal is floating, no supply connection is missing, no ground reference is absent, and no voltmeter lead is reversed. It must also confirm that Vout is not accidentally connected directly to +5V or directly to ground.

A correct scheme should produce a low Vout when the LDR is in darkness and a high Vout when the LDR receives strong light. This behavior proves that the scheme matches the intended design.

3.15 Scheme Image Placement

The scheme image should be inserted after the explanation of the circuit wiring and before the chapter summary. This position is appropriate because the reader first receives the written explanation and then sees the visual confirmation of the complete circuit.

The image should show the final Proteus circuit clearly. It should include V1, LDR1, R1, VM1, GND, Vout, +5V, and the torch. The image should not be overloaded with unnecessary notes. Its main purpose is to show the exact final circuit layout.

The image caption should be formal and precise. A suitable caption is:

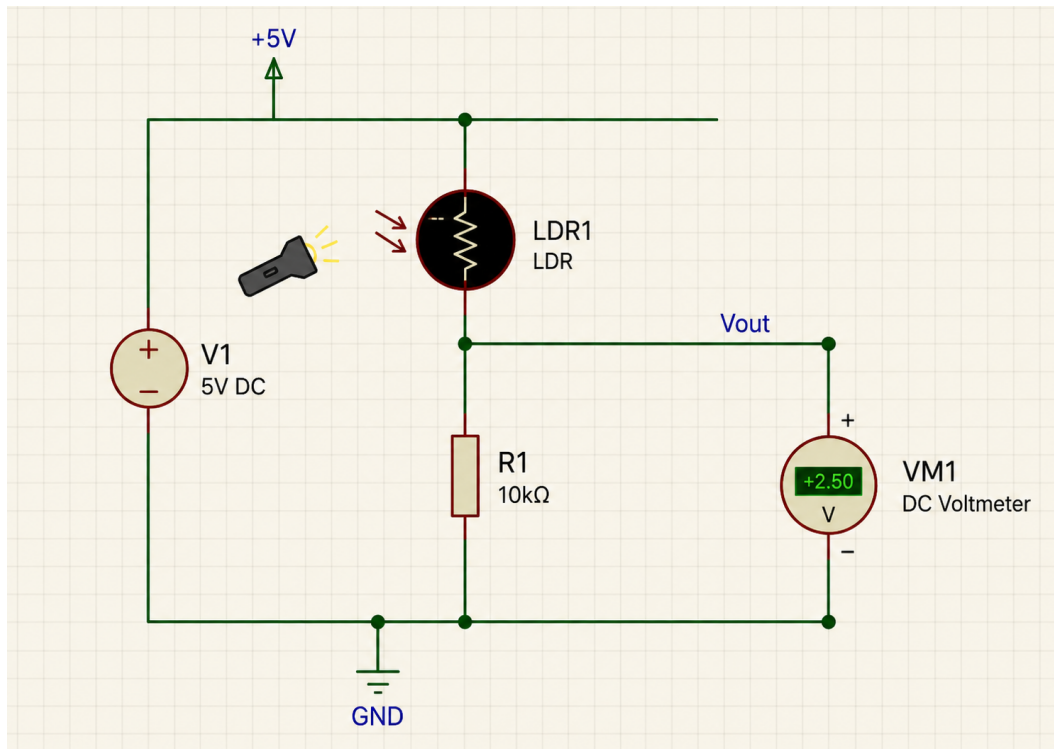


Figure 3.1: Complete Proteus circuit scheme

Figure 3.1. Proteus scheme of the light sensitive voltage divider circuit using an LDR, a 10 kΩ fixed resistor, a 5 V DC supply, and a DC voltmeter connected between Vout and ground.

The image must support the written scheme. It should not replace the explanation. The text gives the technical meaning, while the image gives the visual structure.

3.16 Chapter Summary

This chapter presented the complete Proteus scheme of the light sensitive switch circuit. The scheme used a 5 V DC supply, an LDR, a 10 kΩ fixed resistor, a DC voltmeter, a ground terminal, and a virtual torch. The LDR was connected between the positive supply and the output node, while the fixed resistor was connected between the output node and ground.

The output voltage was defined at the junction between the LDR and the fixed resistor. The voltmeter was connected between this junction and ground in order to measure the voltage produced by the divider. The torch was placed near the LDR to change the simulated light level.

The chapter also established the exact relationship between the circuit wiring and the voltage divider equation. Because the LDR is the upper resistive element and the fixed resistor is the lower resistive element, the output voltage is given by:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This confirms that the circuit scheme is consistent with the expected behavior. When

light intensity increases, the LDR resistance decreases, and the output voltage rises. Therefore, the Proteus scheme correctly implements the sensing stage of the light sensitive switch project.

4

Simulation

4.1 Introduction to Simulation

Simulation is an essential stage in the development of the light sensitive switch circuit because it allows the circuit behavior to be tested before physical construction. In this project, Proteus is used to observe how the output voltage changes when the light intensity applied to the light dependent resistor changes. The simulation confirms whether the theoretical voltage divider calculation and the schematic wiring produce the expected electrical response.

The circuit consists of a five volt direct current supply, one light dependent resistor, one fixed resistor, one direct current voltmeter, one ground terminal, and a virtual torch. The light dependent resistor and the fixed resistor are connected as a voltage divider. The output voltage is measured at the junction between these two resistive elements. The main purpose of the simulation is to verify that the output voltage increases when the virtual torch is moved closer to the light dependent resistor.

This chapter explains the Proteus simulation environment, the required component settings, the measurement method, the simulation procedure, the expected voltage readings, and the comparison between theoretical and simulated results.

4.2 Purpose of Proteus Simulation

The purpose of Proteus simulation is to confirm that the designed circuit performs correctly under different light conditions. The simulation provides a controlled environment where the light intensity can be changed without physically replacing components or rebuilding the circuit. This makes it possible to observe the relationship between light intensity, light dependent resistor resistance, and output voltage.

The simulation has four main purposes. First, it verifies that the circuit is connected correctly. Second, it confirms that the voltmeter is measuring the correct node voltage. Third, it demonstrates that the light dependent resistor responds to the virtual torch. Fourth, it provides simulated voltage values that can be compared with the mathematical results obtained from the voltage divider equation.

The most important expected behavior is that the output voltage must rise when the light intensity increases. This occurs because the light dependent resistor is connected

between the positive supply and the output node. When stronger light reaches the light dependent resistor, its resistance decreases. As a result, the voltage drop across the light dependent resistor decreases, and the output node moves closer to the five volt supply level.

4.3 Simulation Setup

The simulation setup is based on the final circuit scheme created in Proteus. The circuit uses a five volt direct current source as the supply voltage. The positive terminal of the supply is connected to the top terminal of the light dependent resistor. The bottom terminal of the light dependent resistor is connected to the output node. The fixed resistor is connected between the output node and ground. The voltmeter is connected between the output node and ground to measure the output voltage.

The virtual torch is placed near the light dependent resistor. Its position and brightness are adjusted during simulation to represent different light conditions. When the torch is far from the light dependent resistor, the light dependent resistor remains in a high resistance state. When the torch is moved closer, the light dependent resistor enters a lower resistance state.

The simulation setup must contain the following elements:

Table 4.1: *Proteus components required for the circuit*

Component	Label	Setting or Function
Direct current voltage source	V1	5 V
Light dependent resistor	LDR1	Light sensitive resistance
Fixed resistor	R1	10 k Ω
Direct current voltmeter	VM1	Measures output voltage
Ground terminal	GND	Zero volt reference
Virtual torch	Torch	Controls light intensity

The output node must be clearly labelled as (V_{out}). This label is important because the voltmeter reading must be taken exactly from this junction.

4.4 Supply Voltage Setting

The supply voltage is set to five volts direct current. This value is used because it is a common low voltage level in basic electronic circuits and is suitable for demonstrating the behavior of a voltage divider. The positive terminal of the supply is connected to the top of the light dependent resistor, while the negative terminal is connected to ground.

The supply voltage is represented as:

$$V_{CC} = 5V$$

This value is used in every theoretical calculation and must remain constant during simulation. If the supply voltage is changed, the output voltage values will also change. Therefore, the simulation result can only be compared with the theoretical result if the supply voltage is kept at five volts.

The correct supply connection is:

$$\begin{aligned} V1^+ &\rightarrow LDR1_{top} \\ V1^- &\rightarrow GND \end{aligned}$$

If the negative terminal of the supply is not connected to ground, the circuit will not have a proper reference point. In that case, the voltmeter may show an incorrect value or no meaningful value.

4.5 Resistor Value Setting

The fixed resistor is set to ten kilohms. This resistor is connected between the output node and ground. Its function is to form the lower part of the voltage divider. The selected value of ten kilohms is suitable because it provides a clear voltage change when paired with typical light dependent resistor values.

The fixed resistor value is:

$$R_1 = 10k\Omega$$

In this circuit, (R_1) is the bottom resistor in the voltage divider. The light dependent resistor is the top resistor. Therefore, the output voltage is calculated using the bottom resistor value in the numerator of the voltage divider equation.

The circuit specific voltage equation is:

$$V_{out} = V_{CC} \times \frac{R_1}{R_{LDR} + R_1}$$

After substituting the known values:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This equation shows that the output voltage depends directly on the resistance of the light dependent resistor. When (R_{LDR}) decreases, the denominator becomes smaller, and (V_{out}) becomes larger.

4.6 LDR and Torch Configuration

The light dependent resistor is the sensing component of the circuit. Its resistance changes according to the light intensity applied to it. In a dark condition, the resistance of the light dependent resistor is high. In a bright condition, the resistance of the light dependent resistor is low.

The virtual torch in Proteus is used to control the amount of light reaching the light dependent resistor. When the torch is far from the light dependent resistor, the simulated light intensity is low. When the torch is moved closer, the simulated light intensity increases. The circuit response is observed by watching the value displayed on the direct current voltmeter.

The expected relationship is:

$$\begin{aligned} \text{Light} \uparrow &\Rightarrow R_{LDR} \downarrow \\ R_{LDR} \downarrow &\Rightarrow V_{out} \uparrow \end{aligned}$$

This relationship is the central behavior that the simulation must demonstrate. If the output voltage decreases when light increases, then the light dependent resistor and fixed resistor are most likely placed in the opposite positions.

4.7 Voltmeter Measurement Setup

The direct current voltmeter is connected to measure the voltage at the output node. The positive lead of the voltmeter is connected to (V_{out}), and the negative lead is connected to ground. This connection measures the potential difference between the voltage divider junction and the zero volt reference.

The correct voltmeter connection is:

$$\begin{aligned} VM1^+ &\rightarrow V_{out} \\ VM1^- &\rightarrow GND \end{aligned}$$

The voltmeter must not be connected in series with the circuit. It must be connected in parallel with the output measurement point and ground. If the voltmeter leads are reversed, the displayed value may appear as a negative voltage. If the voltmeter is connected to the wrong node, the simulation will not represent the actual output voltage of the voltage divider.

The measured output voltage is:

$$V_{out} = V(V_{junction}) - V(GND)$$

Since ground is zero volts, this becomes:

$$V_{out} = V(V_{junction}) - 0V$$

Therefore:

$$V_{out} = V(V_{junction})$$

4.8 Simulation Procedure

The simulation procedure begins by opening the completed Proteus circuit. The voltage source is checked and set to five volts direct current. The fixed resistor is checked and set to ten kilohms. The light dependent resistor is placed above the output node, and the fixed resistor is placed below the output node. The voltmeter positive lead is connected to the output node, while the voltmeter negative lead is connected to ground.

After confirming the connections, the simulation is started. The first test is performed under a dark or weak light condition by placing the torch far from the light dependent resistor. The voltmeter reading is recorded. The second test is performed under a medium light condition by moving the torch to a moderate distance. The voltmeter reading is recorded again. The third test is performed under a bright light con-

dition by moving the torch close to the light dependent resistor. The final voltmeter reading is recorded.

The procedure must be carried out in the following order:

Open the completed Proteus circuit.

Confirm that the source voltage is five volts.

Confirm that the fixed resistor value is ten kilohms.

Confirm that the light dependent resistor is connected between the positive supply and (V_{out}).

Confirm that the fixed resistor is connected between (V_{out}) and ground.

Confirm that the voltmeter positive lead is connected to (V_{out}).

Confirm that the voltmeter negative lead is connected to ground.

Start the simulation.

Observe the voltage under weak light.

Move the torch to create medium light.

Observe the voltage under medium light.

Move the torch closer to create bright light.

Observe the voltage under bright light.

Compare the simulated readings with theoretical values.

4.9 Dark Condition Simulation

In the dark condition, the torch is placed far from the light dependent resistor or its brightness is reduced. Under this condition, the light dependent resistor has a high resistance. A typical assumed dark resistance is one hundred kilohms.

$$R_{LDR} = 100k\Omega$$

Using the voltage divider equation:

$$\begin{aligned} V_{out} &= 5V \times \frac{10k\Omega}{100k\Omega + 10k\Omega} \\ V_{out} &= 5V \times \frac{10k\Omega}{110k\Omega} \\ V_{out} &= 5V \times 0.0909 \\ V_{out} &\approx 0.45V \end{aligned}$$

The simulation should therefore show a low output voltage in the dark condition. This confirms that a high light dependent resistor resistance causes most of the supply voltage to drop across the light dependent resistor, leaving only a small voltage at the output node.

4.10 Medium Light Simulation

In the medium light condition, the torch is placed at a moderate distance from the light dependent resistor. Under this condition, the resistance of the light dependent resistor may become close to the value of the fixed resistor. A typical assumed medium resistance is ten kilohms.

$$R_{LDR} = 10k\Omega$$

Using the voltage divider equation:

$$\begin{aligned} V_{out} &= 5V \times \frac{10k\Omega}{10k\Omega + 10k\Omega} \\ V_{out} &= 5V \times \frac{10k\Omega}{20k\Omega} \\ V_{out} &= 5V \times 0.5 \\ V_{out} &= 2.5V \end{aligned}$$

The simulation should show an output voltage close to half of the supply voltage when the light dependent resistor and the fixed resistor have equal resistance. This condition is important because it confirms that the voltage divider is balanced when both resistive elements have the same value.

4.11 Bright Light Simulation

In the bright light condition, the torch is moved close to the light dependent resistor or its brightness is increased. Under this condition, the resistance of the light dependent resistor decreases significantly. A typical assumed bright light resistance is one kilohm.

$$R_{LDR} = 1k\Omega$$

Using the voltage divider equation:

$$\begin{aligned} V_{out} &= 5V \times \frac{10k\Omega}{1k\Omega + 10k\Omega} \\ V_{out} &= 5V \times \frac{10k\Omega}{11k\Omega} \\ V_{out} &= 5V \times 0.909 \\ V_{out} &\approx 4.55V \end{aligned}$$

The simulation should therefore show a high output voltage in bright light. This result confirms the intended operation of the circuit. As the torch is moved closer, the light dependent resistor resistance decreases, and the output voltage rises toward the five volt supply level.

4.12 Simulation Observation Table

The simulation results should be recorded in a clear observation table. The table must include the light condition, the assumed or observed light dependent resistor state, the theoretical voltage, the simulated voltage, and the interpretation of the result.

The exact simulated values may vary slightly depending on the Proteus light dependent resistor model, the torch brightness, and the torch distance. However, the trend must remain the same. The output voltage must increase as the light intensity increases.

If actual readings are recorded during simulation, they can be compared with the theoretical values using absolute error:

Table 4.2: Simulation observation table

Light Condition	Assumed LDR Resistance	Theoretical Output Voltage	Expected Simulated Output Voltage	Interpretation
Dark condition	100 k Ω	0.45 V	approximately 0.45 V	Low output voltage
Medium light condition	10 k Ω	2.50 V	approximately 2.50 V	Mid level output voltage
Bright light condition	1 k Ω	4.55 V	approximately 4.55 V	High output voltage

$$\text{Absolute Error} = |V_{\text{simulated}} - V_{\text{theoretical}}|$$

The percentage error can be calculated as:

$$\text{Percentage Error} = \frac{|V_{\text{simulated}} - V_{\text{theoretical}}|}{V_{\text{theoretical}}} \times 100$$

A small difference between theoretical and simulated values is acceptable if the overall trend is correct and the difference is caused by model settings rather than wiring error.

4.13 Comparison Between Theory and Simulation

The theoretical analysis predicts that the output voltage depends on the resistance of the light dependent resistor. The simulation is expected to confirm the same behavior. When the light dependent resistor resistance is high, the output voltage is low. When the light dependent resistor resistance is equal to the fixed resistor value, the output voltage is approximately half of the supply voltage. When the light dependent resistor resistance is low, the output voltage is high.

The theoretical relationship is:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This equation proves that (V_{out}) is controlled by (R_{LDR}). Since (R_{LDR}) decreases when light increases, the output voltage must increase.

The comparison can be expressed as follows:

Table 4.3: Comparison between theoretical prediction and simulation requirement

Condition	Theoretical Prediction	Simulation Requirement
Dark	(R_{LDR}) is high and (V_{out}) is low	Voltmeter must show a low value
Medium light	(R_{LDR}) is near 10 k Ω and (V_{out}) is near 2.5 V	Voltmeter must show a middle value
Bright light	(R_{LDR}) is low and (V_{out}) is high	Voltmeter must show a high value

The simulation supports the theory if the voltmeter reading rises as the torch moves closer to the light dependent resistor. If the simulated result follows this trend, then the circuit is correctly designed and correctly connected.

4.14 Interpretation of Simulation Results

The simulation results show the conversion of light intensity into voltage. The light dependent resistor acts as the sensing element, while the fixed resistor converts the resistance variation into a measurable voltage variation. The voltmeter displays this voltage variation at the output node.

In darkness, the resistance of the light dependent resistor is high. Because it is connected between the positive supply and the output node, a large part of the supply voltage drops across the light dependent resistor. As a result, the output voltage remains low.

Under medium light, the resistance of the light dependent resistor becomes closer to the value of the fixed resistor. In this condition, the voltage divider becomes more balanced, and the output voltage approaches half of the supply voltage.

Under bright light, the resistance of the light dependent resistor becomes low. A smaller voltage drop occurs across the light dependent resistor, and the output node rises toward the positive supply voltage. Therefore, the voltmeter displays a higher value.

The simulation confirms the intended circuit behavior:

$$\begin{array}{l} \text{Light Intensity} \uparrow \\ R_{LDR} \downarrow \\ V_{out} \uparrow \end{array}$$

This result proves that the circuit operates correctly as a light sensitive voltage divider. It also demonstrates the sensing stage of a light sensitive switch, because the output voltage can later be used as an input to a transistor, comparator, or microcontroller threshold circuit.

4.15 Simulation Image Placement

The simulation image should be inserted after the explanation of the simulation results. It should be labelled as Figure 4.1. Proteus Simulation of the Light Sensitive Voltage Divider Circuit.

The image must show the completed Proteus schematic during simulation. It should include the five volt supply, the light dependent resistor, the ten kilohm resistor, the output node, the voltmeter, the ground terminal, and the virtual torch. The torch should be positioned near the light dependent resistor to demonstrate the bright light condition. The voltmeter should display a high output voltage, preferably close to 4.55 V if the light dependent resistor is modeled near one kilohm.

The figure caption should be written as:

Figure 4.1. Proteus simulation of the LDR voltage divider circuit showing an increased output voltage when the torch is moved close to the light dependent resistor.

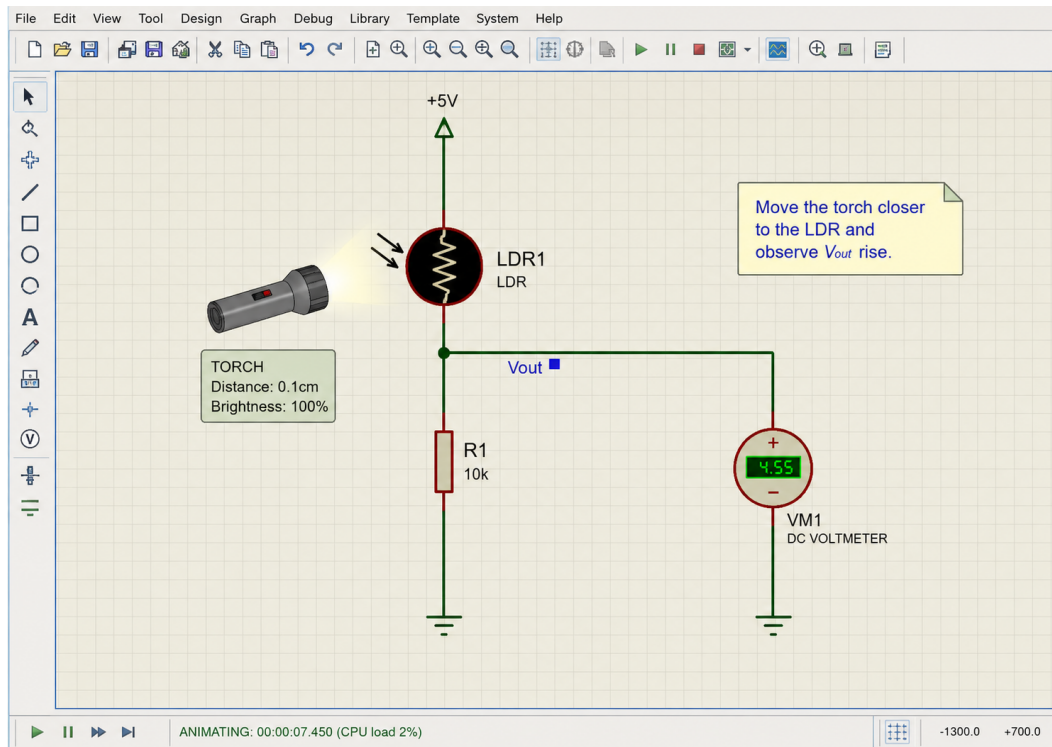


Figure 4.1: Proteus simulation result for bright light condition

The image must support the written explanation. It should not be decorative. It must function as visual evidence that the simulated circuit produces the expected voltage response.

4.16 Chapter Summary

This chapter presented the simulation of the light sensitive switch circuit in Proteus. The simulation was used to verify the behavior of the light dependent resistor voltage divider. The circuit was supplied with five volts direct current, and a ten kilohm fixed resistor was connected between the output node and ground. The light dependent resistor was connected between the positive supply and the output node. A direct current voltmeter was used to measure the voltage at the output node.

The simulation demonstrated that the output voltage changes according to the light intensity applied to the light dependent resistor. In the dark condition, the resistance of the light dependent resistor is high, and the output voltage is low. In the medium light condition, the output voltage approaches the middle of the supply range. In the bright light condition, the resistance of the light dependent resistor becomes low, and the output voltage rises.

The mathematical analysis and the simulated behavior agree with each other. The voltage divider equation shows that the output voltage must increase when the light dependent resistor resistance decreases. The Proteus simulation confirms this result by showing that the voltmeter reading increases when the torch is moved closer to the light dependent resistor.

Therefore, the simulation proves that the designed circuit correctly converts light

variation into voltage variation. This confirms that the circuit is suitable as the sensing stage of a light sensitive switch.

5

Troubleshooting

5.1 Introduction to Troubleshooting

Troubleshooting is an essential part of the light-sensitive switch project because the circuit does not only need to be drawn correctly; it must also behave correctly under changing light conditions. In this project, the expected behavior is that the output voltage at the junction between the LDR and the fixed resistor changes when the light intensity on the LDR changes. Since the LDR is connected between the positive supply and the output node, and the fixed resistor is connected between the output node and ground, the output voltage should increase when the LDR receives stronger light.

The purpose of this chapter is to identify possible faults that may occur during the construction or simulation of the circuit and to explain how each fault can be tested and corrected. The troubleshooting process is based on the circuit structure, the voltage divider equation, the polarity of the voltmeter, the presence of a common ground, and the correct response of the LDR to the virtual torch in Proteus.

The correct theoretical relationship for the circuit is:

$$V_{out} = V_{CC} \times \frac{R_1}{R_{LDR} + R_1}$$

In this project:

$$\begin{aligned} V_{CC} &= 5V \\ R_1 &= 10k\Omega \end{aligned}$$

Therefore:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This formula is the main reference for troubleshooting. If the measured voltage does not follow this relationship, the cause must be found in the supply connection, ground connection, voltmeter connection, LDR behavior, resistor value, or simulation setup.

5.2 Importance of Fault Checking

Fault checking is important because a circuit may appear visually correct while still failing electrically. A wire may not be connected to a node, a ground terminal may be missing, the voltmeter may be connected with reversed polarity, or the LDR may not be receiving light from the torch. These faults can produce incorrect readings even when the schematic looks almost complete.

In this project, fault checking is especially important because the output voltage depends on the relationship between two resistances. The fixed resistor has a constant value, but the LDR changes its resistance according to light intensity. If the LDR does not respond to light, if the resistor value is not suitable, or if the output node is not measured correctly, the circuit will not demonstrate the intended result.

The correct behavior can be summarized as follows:

Light intensity increases
 R_{LDR} decreases
 V_{out} increases

This chain must be preserved. Any result that breaks this chain indicates a fault. Therefore, troubleshooting is not only a repair activity; it is a method of proving that the circuit, calculation, and simulation are consistent with one another.

5.3 Voltmeter Reading is Zero

A zero voltmeter reading means that the voltmeter is measuring no potential difference between its positive and negative terminals. In this project, the voltmeter should measure the voltage between the output node and ground. If the reading remains at zero, the first possibility is that the positive terminal of the voltmeter is not correctly connected to the output node.

The output node is the junction between the bottom terminal of the LDR and the top terminal of the fixed resistor. The positive terminal of the voltmeter must be connected to this junction. The negative terminal of the voltmeter must be connected to ground. If both voltmeter terminals are accidentally connected to ground, the reading will be zero because both terminals are at the same potential.

A zero reading may also occur if the positive supply is not connected to the LDR. In that case, no voltage is applied across the divider, and the output node cannot rise above ground. The correct supply path is:

+ 5V $\rightarrow$ LDR
 $\rightarrow V_{out}$
 $\rightarrow R_1$
 $\rightarrow GND$

If the path from the supply to the LDR is open, the divider does not operate. Another possible cause is that the output node is shorted directly to ground. If V_{out} is unintentionally connected to ground, the voltmeter will correctly show zero, but the circuit will be faulty.

To correct this problem, the following checks must be made. The positive supply must be connected to the top terminal of the LDR. The bottom terminal of the LDR must be connected to the output node. The top terminal of the fixed resistor must be connected to the same output node. The bottom terminal of the fixed resistor must be connected to ground. The voltmeter positive terminal must be connected to V_{out} , and the voltmeter negative terminal must be connected to ground.

5.4 Voltmeter Reading is Negative

A negative voltmeter reading usually means that the voltmeter polarity is reversed. A voltmeter measures the potential at its positive terminal relative to its negative terminal. If the positive lead is connected to ground and the negative lead is connected to V_{out} , the meter will display a negative value.

For this circuit, the correct voltmeter connection is:

$$\begin{aligned} VM1_{positive} &\rightarrow V_{out} \\ VM1_{negative} &\rightarrow GND \end{aligned}$$

If the meter displays a negative voltage, the correction is to swap the voltmeter leads. The positive terminal must be placed at the output node, and the negative terminal must be placed at ground.

A negative reading does not necessarily mean that the voltage divider itself is wrong. It often means that the measuring instrument is connected in the opposite direction. Therefore, before changing the circuit, the polarity of the voltmeter must be checked first.

The expected output voltage should normally remain between zero volts and five volts:

$$0V \leq V_{out} \leq 5V$$

If the supply is five volts and the divider is correctly connected, a large negative reading is not part of the expected circuit behavior. It indicates an incorrect measurement reference or reversed meter polarity.

5.5 Output Voltage Does Not Change

If the output voltage does not change when the torch is moved closer to or farther from the LDR, then the circuit is failing to convert light variation into voltage variation. This problem can have several causes.

The first possible cause is that the torch is not directed toward the LDR. In Proteus, the virtual torch must affect the LDR model. If the torch is not aimed correctly, the LDR resistance may remain unchanged. Since the voltage divider depends on a changing value of (R_{LDR}), a constant LDR resistance produces a constant output voltage.

The second possible cause is that the LDR model is not configured correctly. If the selected LDR component does not respond to the torch or if the simulation model is incomplete, the output voltage may remain fixed. The LDR must be a light-dependent component, not an ordinary fixed resistor.

The third possible cause is incorrect wiring. If the LDR is not connected as the upper resistor in the divider, light changes may not produce the expected output movement. The intended connection is:

$$\begin{aligned} +5V &\rightarrow LDR \\ &\rightarrow V_{out} \\ &\rightarrow R_1 \\ &\rightarrow GND \end{aligned}$$

The fourth possible cause is that the voltmeter is connected to the wrong node. If the voltmeter is not measuring the junction between the LDR and the fixed resistor, the reading may not reflect the divider output.

The mathematical reason for the expected voltage change is:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

When (R_{LDR}) changes, the denominator changes. Therefore, (V_{out}) must also change. If (V_{out}) remains constant, then either (R_{LDR}) is not changing, or the measured node is not the correct output node.

5.6 Output Voltage Changes Too Little

If the output voltage changes only slightly, the circuit is working partially, but its sensitivity is not strong enough for clear observation. This usually happens when the fixed resistor value does not match the useful resistance range of the LDR.

The voltage divider is most sensitive when the fixed resistor has a value near the LDR resistance in the important light range. If (R_1) is much smaller or much larger than the active LDR resistance range, the output voltage may remain close to one extreme and may not move enough to be easily observed.

The sensitivity can be understood from the equation:

$$V_{out} = 5V \times \frac{R_1}{R_{LDR} + R_1}$$

If (R_1) is too small compared with (R_{LDR}), then the fraction becomes small, and the output voltage remains low. If (R_1) is too large compared with (R_{LDR}), then the output voltage may remain near the supply voltage. In both cases, the voltage change may become less useful.

For the default value:

$$R_1 = 10k\Omega$$

the circuit gives a balanced midpoint when:

$$R_{LDR} = 10k\Omega$$

because:

$$V_{out} = 5V \times \frac{10k\Omega}{10k\Omega + 10k\Omega}$$

$$V_{out} = 2.5V$$

If the observed voltage change is too small, the value of (R_1) may be adjusted. A value such as ($22k\Omega$) or ($47k\Omega$) may produce a clearer voltage movement if the LDR resistance remains relatively high under the available light conditions.

5.7 Output Voltage is Always High

If the output voltage remains close to five volts, the output node is staying near the positive supply level. This means that the path from Vout to ground may be weak, disconnected, or incorrectly wired.

In the correct circuit, the fixed resistor provides the path from Vout to ground:

$$V_{out} \rightarrow R_1$$

$$\rightarrow GND$$

If this path is open, the output node may float or remain pulled toward the positive supply through the LDR. The most likely cause is that the bottom terminal of (R_1) is not connected to ground.

Another possible cause is that Vout is accidentally connected directly to the positive supply. In that case, the voltmeter will read approximately five volts regardless of the LDR resistance. This is not voltage divider behavior because the output node is no longer controlled by the ratio between (R_{LDR}) and (R_1).

The correct condition is:

$$V_{out} < 5V$$

unless (R_{LDR}) becomes extremely small compared with (R_1). Even under bright light, the output should approach five volts but should not be treated as directly shorted to five volts.

To correct this problem, the bottom side of (R_1) must be checked for a valid ground connection. The output node must also be checked to ensure that it is not directly connected to the positive rail.

5.8 Output Voltage is Always Low

If the output voltage remains close to zero volts, the output node is staying near ground. This means that the path from the positive supply to Vout may be missing, weak, or incorrectly connected.

In the correct circuit, the LDR provides the path from the positive supply to the output node:

$$+5V \rightarrow LDR$$

$$\rightarrow V_{out}$$

If the LDR is disconnected from the positive supply, the output node cannot rise. The fixed resistor will hold the output node at ground, so the voltmeter will read approximately zero volts.

Another possible cause is that V_{out} is accidentally shorted to ground. If the output node is connected directly to ground, the voltage divider is bypassed, and no meaningful output voltage can be produced.

A third possible cause is that the LDR resistance remains extremely high because the torch is not affecting it. In darkness, a high LDR resistance produces a low output voltage. For example, if:

$$R_{LDR} = 100k\Omega$$

then:

$$V_{out} = 5V \times \frac{10k\Omega}{100k\Omega + 10k\Omega}$$

$$V_{out} \approx 0.45V$$

This value is low, but it is not exactly zero. If the meter shows exactly zero, the fault is more likely to be an open supply path, a ground short, or a measurement error.

5.9 Response is Inverted

The response is inverted when the output voltage decreases as light intensity increases. This behavior is opposite to the intended design of this project.

In the intended circuit, the LDR is the upper element and the fixed resistor is the lower element:

$$\begin{aligned} +5V &\rightarrow LDR \\ &\rightarrow V_{out} \\ &\rightarrow R_1 \\ &\rightarrow GND \end{aligned}$$

This produces:

$$\begin{aligned} \text{More light} &\rightarrow R_{LDR} \text{ decreases} \\ &\rightarrow V_{out} \text{ increases} \end{aligned}$$

If the LDR and fixed resistor are swapped, the circuit becomes:

$$\begin{aligned} +5V &\rightarrow R_1 \\ &\rightarrow V_{out} \\ &\rightarrow LDR \\ &\rightarrow GND \end{aligned}$$

In that arrangement, the output equation changes to:

$$V_{out} = 5V \times \frac{R_{LDR}}{R_1 + R_{LDR}}$$

When light increases, (R_{LDR}) decreases. Since (R_{LDR}) is now in the numerator, the output voltage decreases. This is the mathematical reason for the inverted response.

Therefore, the correction depends on the desired behavior. If the report requires V_{out} to rise with light, the LDR must be placed between +5V and V_{out} , and the fixed resistor must be placed between V_{out} and ground.

5.10 Incorrect Resistor Value Problem

The fixed resistor value strongly affects the output voltage range and sensitivity. In this project, (R_1) is selected as ($10k\Omega$). This value is suitable because many common LDRs pass through a resistance range near ($10k\Omega$) under moderate light conditions, which allows the voltage divider to produce a clear middle output around (2.5V).

If the resistor value is too low, the output voltage may remain low for much of the light range. If the resistor value is too high, the output voltage may remain high for much of the light range. The resistor value does not only affect the amount of current; it also determines how strongly the output voltage responds to changes in LDR resistance.

The circuit equation is:

$$V_{out} = 5V \times \frac{R_1}{R_{LDR} + R_1}$$

This equation shows that changing (R_1) changes the output voltage for every LDR resistance value.

For example, if:

$$R_{LDR} = 10k\Omega$$

and:

$$R_1 = 10k\Omega$$

then:

$$V_{out} = 2.5V$$

If (R_1) is increased while (R_{LDR}) remains the same, the output voltage increases. If (R_1) is decreased while (R_{LDR}) remains the same, the output voltage decreases.

Therefore, an incorrect resistor value can make the circuit appear insensitive or incorrectly biased. The resistor value should be selected according to the expected light range and the required output voltage level.

5.11 Missing Ground Problem

A missing ground is one of the most serious faults in the circuit. Ground provides the zero volt reference for the supply, the resistor, and the voltmeter. Without ground, the circuit has no stable reference point, and the measured voltage may become meaningless.

In the correct circuit, the following points must share the same ground reference:

$$\begin{aligned}
 &V1_{negative} \\
 &R1_{bottom} \\
 &VM1_{negative}
 \end{aligned}$$

If these points are not connected together, the circuit may not operate as a complete voltage divider. The voltmeter may show zero, unstable values, or unexpected readings.

The voltage divider requires a complete path from the positive supply to ground:

$$\begin{aligned}
 &+5V \rightarrow LDR \\
 &\rightarrow V_{out} \\
 &\rightarrow R_1 \\
 &\rightarrow GND
 \end{aligned}$$

If ground is missing, this path is incomplete. The circuit may look connected at the top, but the bottom reference is absent. Therefore, the first correction for unstable or meaningless readings is to verify that the negative terminal of the supply, the bottom terminal of the fixed resistor, and the negative lead of the voltmeter are all connected to the same ground node.

A correct ground connection also ensures that the voltmeter reading represents the true output voltage:

$$V_{out} = V_{junction} - V_{ground}$$

Since:

$$V_{ground} = 0V$$

the voltmeter displays the actual junction voltage.

5.12 Incorrect Torch or LDR Setup

The torch and LDR setup is essential because the simulated light source must affect the resistance of the LDR. If the torch is placed incorrectly, if its brightness is too low, or if it is too far from the LDR, the LDR resistance may not change enough to produce a visible voltage variation.

The LDR must be the active light-sensitive component in the simulation. It should not be replaced by a fixed resistor, because a fixed resistor cannot respond to light. The torch must be directed toward the LDR so that changing the torch distance or brightness changes (R_{LDR}).

The expected relationship is:

$$\begin{aligned}
 &\text{Torch closer to LDR} \rightarrow \text{higher light intensity} \\
 &\text{Higher light intensity} \rightarrow R_{LDR} \text{ decreases} \\
 &R_{LDR} \text{ decreases} \rightarrow V_{out} \text{ increases}
 \end{aligned}$$

If this sequence is not observed, the torch and LDR settings must be checked. The user should confirm that the torch is active, directed toward the LDR, and adjusted dur-

ing simulation. The user should also confirm that the selected LDR component supports light interaction in Proteus.

5.13 Troubleshooting Checklist

The troubleshooting checklist provides a direct method for verifying the circuit. Each item should be checked in order because earlier faults can hide later faults.

First, confirm that the supply voltage is five volts. The source must provide a stable direct current voltage of (5V). If the supply is wrong, every voltage reading in the circuit will also be wrong.

Second, confirm that the LDR is connected between the positive supply and the output node. The top terminal of the LDR must connect to +5V, and the bottom terminal must connect to Vout.

Third, confirm that the fixed resistor is connected between Vout and ground. The top terminal of (R_1) must connect to Vout, and the bottom terminal must connect to ground.

Fourth, confirm that the voltmeter is connected correctly. The positive lead must connect to Vout, and the negative lead must connect to ground.

Fifth, confirm that all ground points are common. The negative terminal of the supply, the bottom terminal of (R_1), and the negative lead of the voltmeter must be connected to the same reference node.

Sixth, confirm that the torch is aimed at the LDR. The torch distance and brightness should be changed during simulation to observe the output response.

Seventh, confirm that the output voltage follows the theoretical trend. In darkness, Vout should be low. In bright light, Vout should be high.

The correct diagnostic conclusion is:

If light increases and V_{out} increases, the circuit behavior is correct.

If light increases and V_{out} decreases, the divider orientation is reversed.

If light changes and V_{out} does not change,
the LDR response or measurement path is faulty.

5.14 Fault-Correction Table

5.15 Troubleshooting Image Placement

The troubleshooting image should be placed after the fault explanation sections and before the chapter summary. Its purpose is to provide visual confirmation of the most important diagnostic points. The image must show the same circuit used in the scheme and simulation chapters so that the reader can compare the correct wiring with the fault-checking instructions.

The image should include the following elements. The LDR must be shown between +5V and Vout. The fixed resistor must be shown between Vout and ground. The voltmeter positive terminal must be shown connected to Vout, and the voltmeter negative terminal must be shown connected to ground. The torch must be shown facing the LDR.

Table 5.1: *Fault-correction table*

Fault condition	Probable cause	Technical explanation	Corrective action
Voltmeter reads zero	Vout is not connected to the voltmeter positive terminal	The meter is not measuring the junction voltage	Connect the positive terminal of VM1 to Vout
Voltmeter reads zero	Ground path is missing	The voltage divider path is incomplete	Connect V1 negative, R1 bottom, and VM1 negative to ground
Voltmeter reads negative	Voltmeter polarity is reversed	The meter is measuring ground relative to Vout	Connect VM1 positive to Vout and VM1 negative to ground
Output voltage does not change	Torch is not affecting the LDR	(R_{LDR}) remains constant	Aim the torch directly at the LDR and vary its distance or brightness
Output voltage does not change	Wrong LDR model is used	The component does not respond to simulated light	Select a light-dependent resistor model that supports torch interaction
Output voltage changes too little	Fixed resistor value is unsuitable	The divider is not sensitive in the selected resistance range	Try $(22k\Omega)$ or $(47k\Omega)$, then compare the voltage range
Output voltage is always high	R1 is not connected to ground	Vout has no proper pull-down path	Connect the bottom terminal of R1 to ground
Output voltage is always high	Vout is shorted to +5V	The divider is bypassed	Remove the direct connection between Vout and +5V
Output voltage is always low	LDR is not connected to +5V	Vout has no proper pull-up path	Connect the top terminal of the LDR to +5V
Output voltage is always low	Vout is shorted to ground	The output node is forced to zero volts	Remove the unwanted ground short
Response is inverted	LDR and R1 positions are swapped	The LDR is in the lower divider position	Place the LDR above Vout and place R1 below Vout
Reading is unstable	Ground reference is not common	The circuit has no single voltage reference	Use one shared ground node for supply, resistor, and meter

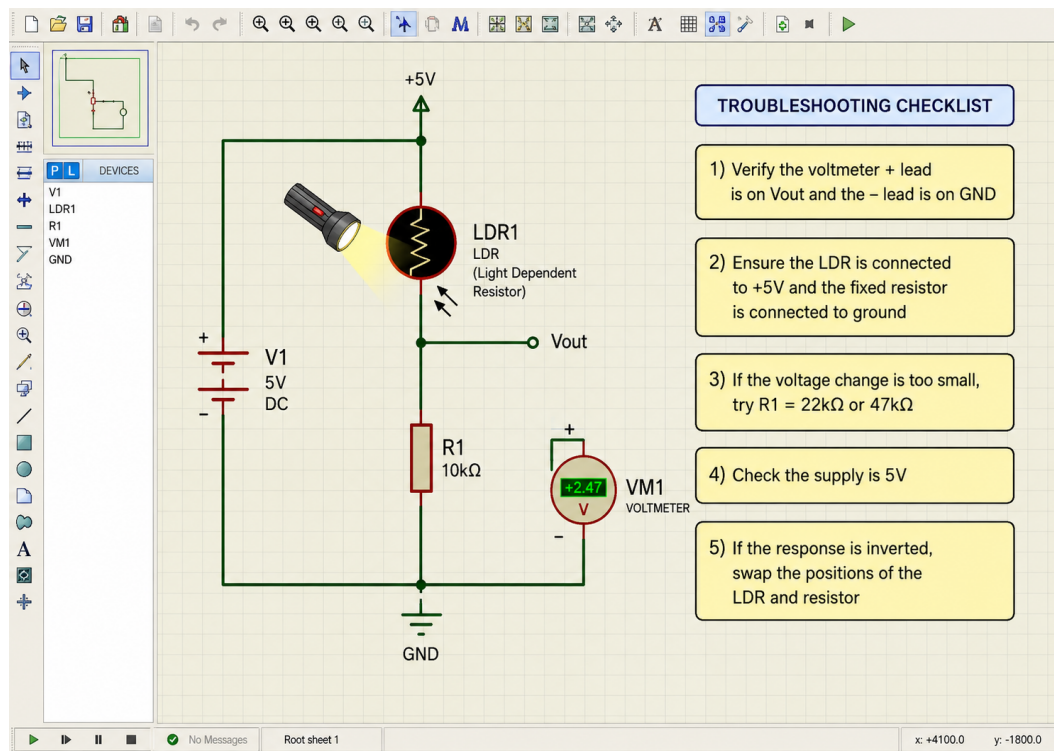


Figure 5.1: Troubleshooting checklist for the light-sensitive switch circuit

The ground node must be clearly visible and connected to the supply negative terminal, resistor bottom terminal, and voltmeter negative terminal.

The image should also contain short fault-checking labels. These labels should state that the supply must be five volts, the voltmeter polarity must be correct, the LDR must be placed above the output node, the fixed resistor must be connected to ground, and the torch must illuminate the LDR.

The troubleshooting image should not replace the written explanation. It should support the explanation by showing the correct physical and electrical placement of the main diagnostic points.

5.16 Chapter Summary

This chapter explained the main faults that may occur in the light-sensitive switch circuit and described how each fault can be diagnosed and corrected. The most important requirement is that the circuit must preserve the correct voltage divider structure. The LDR must be connected between the positive supply and the output node, while the fixed resistor must be connected between the output node and ground.

The output voltage is governed by:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This equation proves the expected behavior of the circuit. When light intensity increases, the resistance of the LDR decreases. Since the LDR resistance is in the denominator, a decrease in (R_{LDR}) causes the output voltage to increase. Therefore, the correct

simulation result is a rising V_{out} when the torch is moved closer to the LDR.

The chapter also showed that most faults can be classified into a small number of categories: incorrect measurement, missing ground, wrong divider orientation, unsuitable resistor value, poor torch alignment, incorrect LDR model, open connection, or short circuit. By checking these categories systematically, the user can locate the fault without guessing.

A properly functioning circuit must satisfy all of the following conditions:

$$V1 = 5V$$

LDR connected from $+5V$ to V_{out}

$R_1 = 10k\Omega$ connected from V_{out} to GND

$VM1_{positive}$ connected to V_{out}

$VM1_{negative}$ connected to GND

Torch directed toward the LDR

When these conditions are satisfied, the circuit correctly demonstrates the sensing stage of a light-sensitive switch.

6

Questions

6.1 Introduction to Project Questions

This chapter presents the main academic questions related to the light-sensitive switch circuit designed using a Light-Dependent Resistor and a voltage divider. The purpose of this chapter is to evaluate the understanding of the project from conceptual, mathematical, practical, and interpretive perspectives. The questions are organized according to the logical structure of the project, beginning with basic concepts and progressing toward circuit calculation, wiring verification, Proteus simulation, troubleshooting, result interpretation, and final project defense.

The chapter also serves as a revision and validation section. It confirms whether the working principle of the circuit has been understood correctly and whether the relationship between light intensity, LDR resistance, and output voltage has been explained with sufficient technical accuracy. Since the circuit depends on a voltage divider arrangement, the questions focus not only on component identification but also on the mathematical reason why the output voltage changes.

The most important idea examined in this chapter is that the LDR does not directly generate voltage. Instead, it changes its resistance when light intensity changes. The voltage divider then converts this resistance variation into a measurable output voltage. Therefore, every question in this chapter supports the same central principle: the circuit output is determined by the position of the LDR, the value of the fixed resistor, the supply voltage, and the voltage divider equation.

6.2 Basic Concept Questions

The basic concept questions are intended to confirm the general purpose and operating idea of the project. These questions test whether the project can be explained clearly without immediately entering detailed calculations or simulation settings.

Question 1. What is the main purpose of this project?

The main purpose of this project is to design and simulate a light-sensitive circuit that changes its output voltage according to the amount of light falling on an LDR. The circuit demonstrates how environmental light can be converted into an electrical voltage signal by using a voltage divider.

Question 2. Why is this circuit called light-sensitive?

The circuit is called light-sensitive because one of its components, the LDR, changes its resistance when the light intensity changes. Since the LDR is part of the voltage divider, this resistance change causes the output voltage to change.

Question 3. What is the main output of the circuit?

The main output of the circuit is the voltage measured at the junction between the LDR and the fixed resistor. This voltage is called (V_{out}).

Question 4. What does the circuit demonstrate?

The circuit demonstrates the conversion of a physical environmental change, namely light intensity, into an electrical quantity, namely voltage. It also demonstrates the practical use of the voltage divider principle.

Question 5. Is the circuit a complete automatic switching system?

The circuit is a sensing stage for a light-sensitive switch. It produces a variable voltage output, but a complete switching system would normally require an additional stage such as a comparator, transistor, relay, LED driver, or microcontroller input.

6.3 LDR Questions

The LDR questions examine the behavior and function of the Light-Dependent Resistor inside the circuit. These questions are essential because the LDR is the sensing element of the project.

Question 1. What is an LDR?

An LDR is a Light-Dependent Resistor. It is a passive electronic component whose resistance changes according to the intensity of light falling on its surface.

Question 2. What happens to the resistance of an LDR in darkness?

In darkness, the resistance of the LDR becomes high. This high resistance limits the current through the upper part of the voltage divider and causes the output voltage to become low in the circuit arrangement used in this project.

Question 3. What happens to the resistance of an LDR in bright light?

In bright light, the resistance of the LDR becomes low. Since the LDR is connected between the positive supply and the output node, this lower resistance allows the output node to move closer to the positive supply voltage.

Question 4. Why does the LDR not produce the output voltage by itself?

The LDR does not produce voltage because it is a resistive component, not a voltage source. It only changes resistance. The output voltage appears because the LDR is connected with a fixed resistor in a voltage divider circuit.

Question 5. Why is the LDR placed on the upper side of the voltage divider in this project?

The LDR is placed on the upper side of the voltage divider so that the output voltage increases when light intensity increases. When the LDR resistance decreases under stronger light, the voltage drop across the LDR decreases, and the output node rises toward the supply voltage.

6.4 Voltage Divider Questions

The voltage divider questions examine the theoretical foundation of the project. These questions prove that the output voltage is not random but follows a definite mathematical relationship.

Question 1. What is a voltage divider?

A voltage divider is a circuit made from two resistive elements connected in series across a voltage supply. The output voltage is taken from the junction between the two resistive elements.

Question 2. Why is a voltage divider used in this project?

A voltage divider is used because it converts the resistance variation of the LDR into a voltage variation. Without the voltage divider, the resistance change of the LDR would not directly provide a useful voltage output.

Question 3. What are the two resistive elements in this project?

The two resistive elements are the LDR and the fixed resistor (R_1). The LDR acts as the variable upper resistor, and (R_1) acts as the fixed lower resistor.

Question 4. Where is the output voltage measured?

The output voltage is measured at the junction between the LDR and (R_1). This point is labeled as (V_{out}).

Question 5. What is the general voltage divider formula?

The general voltage divider formula is:

$$V_{out} = V_{CC} \times \frac{R_{bottom}}{R_{top} + R_{bottom}}$$

In this project, the upper resistance is the LDR resistance, and the lower resistance is the fixed resistor.

Question 6. What is the project-specific voltage divider formula?

Since the LDR is connected at the top and the fixed resistor is connected at the bottom, the formula becomes:

$$V_{out} = 5V \times \frac{R_1}{R_{LDR} + R_1}$$

Since ($R_1 = 10k\Omega$), the formula becomes:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

6.5 Calculation Questions

The calculation questions verify whether the numerical behavior of the circuit is understood. These questions connect the theoretical formula to actual expected voltage values.

Question 1. What is the output voltage when the LDR resistance is ($100k\Omega$)?

Using the formula:

$$V_{out} = 5V \times \frac{10k\Omega}{100k\Omega + 10k\Omega}$$

$$V_{out} = 5V \times \frac{10}{110}$$

$$V_{out} \approx 0.45V$$

Therefore, when the LDR resistance is ($100k\Omega$), the output voltage is approximately ($0.45V$). This represents a dark condition.

Question 2. What is the output voltage when the LDR resistance is ($10k\Omega$)?

Using the formula:

$$V_{out} = 5V \times \frac{10k\Omega}{10k\Omega + 10k\Omega}$$

$$V_{out} = 5V \times \frac{10}{20}$$

$$V_{out} = 2.5V$$

Therefore, when the LDR resistance is equal to the fixed resistor, the output voltage is half of the supply voltage.

Question 3. What is the output voltage when the LDR resistance is ($1k\Omega$)?

Using the formula:

$$V_{out} = 5V \times \frac{10k\Omega}{1k\Omega + 10k\Omega}$$

$$V_{out} = 5V \times \frac{10}{11}$$

$$V_{out} \approx 4.55V$$

Therefore, when the LDR resistance is ($1k\Omega$), the output voltage is approximately ($4.55V$). This represents a bright light condition.

Question 4. Why does (V_{out}) increase when light increases?

When light increases, (R_{LDR}) decreases. In the formula

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

the value of (R_{LDR}) is part of the denominator. When (R_{LDR}) becomes smaller, the denominator becomes smaller, and the value of (V_{out}) becomes larger.

Question 5. What is the relationship between light intensity and output voltage in this circuit?

In this circuit arrangement, light intensity and output voltage have a direct relationship. As light intensity increases, the output voltage increases. As light intensity decreases, the output voltage decreases.

6.6 Scheme and Wiring Questions

The scheme and wiring questions verify whether the circuit can be physically and logically connected without ambiguity. This section is important because even a correct calculation becomes useless if the circuit is wired incorrectly.

Question 1. How is the positive supply connected?

The positive terminal of the (5V) DC supply is connected to the top terminal of the LDR.

Question 2. How is the LDR connected?

The top terminal of the LDR is connected to (+5V). The bottom terminal of the LDR is connected to the output node (V_{out}).

Question 3. How is the fixed resistor connected?

The top terminal of the fixed resistor (R_1) is connected to (V_{out}). The bottom terminal of (R_1) is connected to ground.

Question 4. How is the voltmeter connected?

The positive lead of the voltmeter is connected to (V_{out}). The negative lead of the voltmeter is connected to ground.

Question 5. Why must the ground connection be included?

The ground connection is required because the circuit needs a reference point for voltage measurement. Without ground, the voltmeter reading may become undefined or incorrect.

Question 6. What is the correct wiring order from the supply to ground?

The correct wiring order is:

$$\begin{aligned} &+5V \rightarrow LDR \\ &\rightarrow V_{out} \\ &\rightarrow R_1 \\ &\rightarrow GND \end{aligned}$$

This order must be preserved if the output voltage is expected to rise under stronger light.

Question 7. What happens if the LDR and fixed resistor are swapped?

If the LDR and fixed resistor are swapped, the output response becomes inverted. In that case, the output voltage decreases when the light intensity increases.

6.7 Proteus Simulation Questions

The Proteus simulation questions examine the practical verification of the circuit. This section confirms whether the theoretical behavior can be observed through simulation.

Question 1. Why is Proteus used in this project?

Proteus is used to design and simulate the circuit before physical construction. It allows the output voltage to be measured and the effect of light on the LDR to be observed virtually.

Question 2. Which components are required in Proteus?

The required Proteus components are a (5V) DC voltage source, an LDR, a (10k Ω) resistor, a DC voltmeter, a ground terminal, and a virtual torch.

Question 3. What should be checked before starting the simulation?

Before starting the simulation, the supply voltage must be checked, the resistor value must be confirmed, the LDR must be connected correctly, the voltmeter must be connected from (V_{out}) to ground, and the torch must be aimed at the LDR.

Question 4. What should happen when the torch is moved closer to the LDR?

When the torch is moved closer to the LDR, the light intensity increases. The resistance of the LDR decreases, and the output voltage rises.

Question 5. What should happen when the torch is moved away from the LDR?

When the torch is moved away from the LDR, the light intensity decreases. The resistance of the LDR increases, and the output voltage falls.

Question 6. What does the voltmeter prove in the simulation?

The voltmeter proves that the output voltage changes according to the light condition. It also confirms that the circuit behaves according to the voltage divider equation.

6.8 Troubleshooting Questions

The troubleshooting questions verify whether faults can be identified and corrected logically. This section is necessary because circuit simulation and real circuit construction often fail because of small connection errors.

Question 1. What should be checked if the voltmeter shows zero volts?

If the voltmeter shows zero volts, the first checks should be the voltmeter connection, the ground connection, the supply connection, and the continuity of the (V_{out}) node.

Question 2. What should be checked if the voltmeter shows a negative value?

If the voltmeter shows a negative value, the voltmeter leads are most likely reversed. The positive lead should be connected to (V_{out}), and the negative lead should be connected to ground.

Question 3. What should be checked if the output voltage does not change when the torch moves?

If the output voltage does not change, the torch position, LDR model, LDR connection, simulation settings, and resistor connection should be checked.

Question 4. What should be checked if the output voltage is always high?

If the output voltage is always high, the fixed resistor may not be properly connected to ground, or the output node may be connected directly to the positive supply. The connection from (V_{out}) through (R_1) to ground must be verified.

Question 5. What should be checked if the output voltage is always low?

If the output voltage is always low, the LDR may not be connected to (+5V), or the output node may be shorted to ground. The connection from (+5V) through the LDR to (V_{out}) must be verified.

Question 6. What should be done if the voltage change is too small?

If the voltage change is too small, the value of the fixed resistor should be adjusted. A value such as ($22k\Omega$) or ($47k\Omega$) may produce a more visible voltage change depending on the resistance range of the LDR.

Question 7. What should be done if the circuit response is inverted?

If the output voltage decreases when light increases, the LDR and fixed resistor have probably been placed in the opposite positions. The LDR should be connected between (+5V) and (V_{out}), while the fixed resistor should be connected between (V_{out}) and ground.

6.9 Result Interpretation Questions

The result interpretation questions examine whether the student can explain the meaning of the measured values. This section is important because numerical results must be connected to physical circuit behavior.

Question 1. What does a low (V_{out}) value indicate?

A low (V_{out}) value indicates that the LDR resistance is high. This usually corresponds to a low light or dark condition.

Question 2. What does a high (V_{out}) value indicate?

A high (V_{out}) value indicates that the LDR resistance is low. This usually corresponds to a bright light condition.

Question 3. Why is (2.5V) an important reference value in this project?

The value (2.5V) is important because it occurs when the LDR resistance is equal to the fixed resistor value of ($10k\Omega$). At this point, the supply voltage is divided equally between the LDR and the fixed resistor.

Question 4. How do the calculation and simulation results support each other?

The calculation predicts the expected output voltage for different LDR resistance values, while the simulation shows the practical behavior of the circuit under changing light conditions. If both show that (V_{out}) rises when light increases, then the theoretical and simulated results agree.

Question 5. What is the final meaning of the project result?

The final result means that the circuit successfully converts light intensity into a measurable voltage change. This confirms the function of the LDR voltage divider as the sensing stage of a light-sensitive switch.

6.10 Viva or Oral Defense Questions

The viva or oral defense questions prepare the student to explain the project clearly in spoken academic assessment. The answers should be direct, technically correct, and supported by the circuit principle.

Question 1. Explain the project in one complete sentence.

This project designs and simulates a light-sensitive voltage divider circuit in which an LDR and a fixed resistor produce an output voltage that increases when light intensity increases.

Question 2. What is the most important principle used in the circuit?

The most important principle is the voltage divider principle, because it explains how the resistance change of the LDR becomes a voltage change at the output node.

Question 3. Why does the project use a ($10k\Omega$) resistor?

The ($10k\Omega$) resistor is used as a suitable fixed resistor for forming a voltage divider with the LDR. It gives clear output voltage variation when the LDR resistance changes across common dark and bright conditions.

Question 4. Why is the output voltage not constant?

The output voltage is not constant because the resistance of the LDR changes with light intensity. Since the LDR is part of the voltage divider, its resistance change changes the voltage at the junction node.

Question 5. What would you add to make this a complete automatic switch?

To make the circuit a complete automatic switch, an output stage should be added. This may be a comparator, transistor driver, relay, LED driver, or microcontroller threshold input.

Question 6. What is the strongest evidence that the circuit works correctly?

The strongest evidence is the agreement between the voltage divider calculation and the Proteus simulation result. Both show that (V_{out}) increases when light intensity increases.

6.11 Short Answer Questions

The short answer questions are designed for quick assessment of essential knowledge. Each answer should be concise but technically exact.

Question 1. What does LDR stand for?

LDR stands for Light-Dependent Resistor.

Question 2. What happens to an LDR in bright light?

Its resistance decreases.

Question 3. What happens to an LDR in darkness?

Its resistance increases.

Question 4. What is the supply voltage used in the project?

The supply voltage is (5V) DC.

Question 5. What is the value of the fixed resistor?

The fixed resistor value is ($10k\Omega$).

Question 6. Where is (V_{out}) taken from?

(V_{out}) is taken from the junction between the LDR and (R_1).

Question 7. What instrument measures (V_{out})?

A DC voltmeter measures (V_{out}).

Question 8. What happens to (V_{out}) when light increases?

(V_{out}) increases.

Question 9. What happens to (V_{out}) when light decreases?

(V_{out}) decreases.

Question 10. What is the ground used for?

Ground provides the reference point for voltage measurement.

6.12 Long Answer Questions

The long answer questions require deeper explanation and should be answered in complete academic paragraphs. These questions are suitable for written assessment, report discussion, or final project evaluation.

Question 1. Explain the complete working principle of the circuit.

The circuit works by using an LDR and a fixed resistor as a voltage divider. The LDR is connected between the (5V) supply and the output node, while the fixed resistor is connected between the output node and ground. When light falls on the LDR, its resistance changes. In darkness, the LDR resistance is high, so the output voltage remains low. In bright light, the LDR resistance becomes low, so the output voltage rises. The voltmeter measures this output voltage at the junction between the LDR and the fixed resistor. Therefore, the circuit converts light intensity into a changing voltage signal.

Question 2. Explain mathematically why the output voltage rises in bright light. The output voltage is determined by the voltage divider equation:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

In this equation, (R_{LDR}) is in the denominator. When the light intensity increases, the resistance of the LDR decreases. This reduces the total value of the denominator. Since the numerator remains constant, the value of (V_{out}) increases. Therefore, the mathematical relationship proves that brighter light produces a higher output voltage in this circuit.

Question 3. Explain why the placement of the LDR is important.

The placement of the LDR determines whether the output voltage increases or decreases with light. In this project, the LDR is connected between (+5V) and (V_{out}), while the fixed resistor is connected between (V_{out}) and ground. This arrangement causes the output voltage to rise when the LDR resistance falls. If the LDR and fixed resistor were swapped, the output voltage would fall when light increases. Therefore, the physical position of the LDR in the divider directly controls the direction of the output response.

Question 4. Explain how Proteus simulation verifies the circuit.

Proteus simulation verifies the circuit by allowing the user to observe the output voltage while changing the light falling on the LDR. When the virtual torch is moved closer to the LDR, the voltmeter should show an increase in (V_{out}). When the torch is moved away, the voltmeter should show a decrease in (V_{out}). This simulated behavior confirms that the circuit follows the expected voltage divider principle. The simulation also helps detect wiring errors before building the circuit physically.

Question 5. Explain the main troubleshooting process for this circuit.

The troubleshooting process begins by checking the power supply and ground connections. The (5V) supply must be connected to the top terminal of the LDR, and the negative supply terminal must be connected to ground. Next, the output node must be checked to ensure that it is located between the LDR and the fixed resistor. The voltmeter positive lead must be connected to (V_{out}), and the negative lead must be connected to ground. If the voltage does not change with light, the LDR model, torch position, and resistor value must be checked. If the output response is inverted, the LDR and fixed resistor positions must be corrected.

6.13 Final Review Questions

The final review questions summarize the full project and confirm whether the report has achieved complete technical clarity.

Question 1. What is the central principle of the project?

The central principle is that the LDR changes resistance with light, and the voltage divider converts that resistance change into a measurable output voltage.

Question 2. What is the most important formula in the project?

The most important formula is:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This formula explains the complete relationship between LDR resistance and output voltage.

Question 3. What is the expected result in darkness?

In darkness, the LDR resistance is high, so the output voltage is low.

Question 4. What is the expected result in bright light?

In bright light, the LDR resistance is low, so the output voltage is high.

Question 5. What must be true for the circuit to work correctly?

The LDR must be connected between (+5V) and (V_{out}), the fixed resistor must be connected between (V_{out}) and ground, the voltmeter must measure from (V_{out}) to ground, and the torch must affect the LDR.

Question 6. What is the final conclusion of the project?

The final conclusion is that the LDR voltage divider successfully produces an output voltage that depends on light intensity. The circuit therefore demonstrates the sensing foundation of a light-sensitive switching system.

6.14 Chapter Summary

This chapter presented the complete question structure required to evaluate the light-sensitive switch project. The questions were arranged from basic understanding to advanced explanation, beginning with the purpose of the project and continuing through LDR behavior, voltage divider theory, numerical calculation, circuit wiring, Proteus simulation, troubleshooting, result interpretation, and oral defense preparation.

The chapter confirmed that the circuit operates according to a clear electrical principle. The LDR changes resistance when light intensity changes, and the voltage divider converts that resistance change into the output voltage (V_{out}). In the selected arrangement, the LDR is connected above the output node and the fixed resistor is connected below the output node. As a result, stronger light decreases the LDR resistance and increases the output voltage.

The mathematical foundation of the project is expressed by the formula:

$$V_{out} = 5V \times \frac{10k\Omega}{R_{LDR} + 10k\Omega}$$

This formula explains why the output voltage is low in darkness, approximately half of the supply voltage when the LDR resistance equals ($10k\Omega$), and high under bright light. The Proteus simulation is expected to support this relationship by showing a rise in voltmeter reading when the torch is moved closer to the LDR.

Overall, this chapter strengthens the academic quality of the report by proving that the project can be explained, calculated, simulated, verified, and defended through precise technical reasoning.

